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A Differential Approach to Photometric Stereo with Perspective Viewing*

Roberto Mecca[†‡], Ariel Tankus[§], Aaron Wetzler[†] and Alfred M. Bruckstein[†]

Abstract. Shape from Shading and Photometric Stereo are two fundamental problems in Computer Vision aimed at reconstructing surface depth given either a single image taken under a known light source or multiple images taken under different illuminations from the same viewing angle. Whereas the former uses partial differential equation (PDE) techniques to solve the image irradiance equation, the latter can be expressed as a linear system of equations in surface derivatives when 3 or more images are given. Therefore, it seems that current photometric stereo techniques do not extract all possible depth information from each image by itself. Extending our previous results on this problem, we consider the more realistic perspective projection of surfaces during the photographic process. Under this assumption, there is a unique weak solution (Lipschitz continuous) to the problem at hand, solving the well known convex/concave ambiguity of the Shape from Shading problem. The main contribution of this paper is based on a new differential approach for multi-image Photometric Stereo. Most of the existing works on this topic do not directly address this problem. The common approach is to estimate the gradient field of the surface by minimizing some functional and integrate it afterwards to find the depth and hence the geometry of the object. Our new differential approach allows to solve the problem directly, whilst dealing with images having missing parts. The mathematical well-posedness of the new formulation allows fast numerical algorithm based on a combination of fast-marching and fast-sweeping methods.

Key words. Perspective Shape from Shading, Photometric Stereo, non-linear PDEs system, unique weak Lipschitz solution, finite difference upwind schemes, semi-Lagrangian schemes.

AMS subject classifications. 68T45, 35A02, 65N21, 65N12, 65M25.

1. Introduction.

Reconstruction of three dimensional surfaces from images is one of the most fundamental problems in Computer Vision. Two reconstruction approaches, both of which date back to the 1970s, are Shape from Shading (SfS) [16, 7, 6] and Photometric Stereo [41, 42]. Shape from Shading is aimed at solving the image irradiance equation, which relates the reflectance map to image intensity. Photometric Stereo is a monocular 3D shape reconstruction method based on several images of a scene taken from an identical viewpoint under different illumination conditions. The most common approach in the field divides the task into two steps: recovery of surface gradients first and integration of the resultant gradient field to determine the 3D surface itself later. The goal of the first part is to solve a system of image irradiance equations. When given 3 or more images, this system becomes linear. As such, the gradient field can be recovered analytically. For this reason, Shape from Shading and Photometric Stereo have very diverse methodologies, even though the latter is a generalization of the former.

Recent research in the field of Shape from Shading focused on advanced propagation and partial differential equation (PDE) methods. Among them are the method of level sets of Bruckstein [6] and Kimmel Bruckstein [7], the Hamilton-Jacobi based method of Rouy and

*Research partly supported by Broadcom foundation
[†]Faculty of Computer Science, Technion University, Haifa 32000, Israel
[‡]Pattern Analysis & Computer Vision, Istituto Italiano di Tecnologia, via Morego, 30, 16163 Genova
[§]Tel Aviv Sourasky Medical Center, Department of Neurosurgery and Department of Neurology

Tourin [30], later extended by Prados et al. [28], a control-theory based algorithm of Dupuis and Oliensis [25], and a finite element method by Falcone and Sagona [13]. A single-pass very efficient algorithm is the Fast Marching method [18], see [12] for a recent survey.

A more recent development in the field of Shape from Shading is the transition from the assumption of an orthographic projection of the photographed surface onto the image plane to an assumption of a perspective projection [39, 36, 37, 38, 27, 29].

Photometric Stereo research has focused on surface reconstruction from three or more images (see [1] for a review). We denote the problem with n images by PS_n. Conditions on the illumination and surface reflectance required to obtain uniqueness of solution for three light source Photometric Stereo, are described by Okatani and Deguchi [24]. Even when the light source intensity and directions are unknown, Shashua [32] has shown that three or more images provide enough information to determine the scaled surface normals of an object up to an unknown linear transformation, which allows the reconstruction of the surface also under unknown lighting conditions (assuming distant light sources) [3].

Most of the studies which addressed the Shape from Photometric Stereo problem using multiple images employ a two-step approach [34]:

1. Estimation of the first derivatives of the surface (usually via some minimization algorithms);
2. Recovery of the height from the gradient field, see e.g. [11, 10, 2], (by integration or by functional minimisation).

With the aim to compute the height of the surface directly, several variational approaches have been studied using the Euler-Lagrange equations associated to functionals that involve the given images and iterate based on some prior information on the surface. Such PDEs have several drawbacks, most important are the high order of the partial derivatives involved and the impossibility to have solution with regularity less than a C^2 function [17].

Due to the analytic solution for PS_3, only few studies investigated 2-image Photometric Stereo, PS_2 (for example, [26, 22, 20, 40]). A comprehensive work on existence and uniqueness in PS_2 is that of Kozera [19] that retraces the study of Onn and Bruckstein [26]. In PS_2, [26, 21] showed it is impossible to recover the gradient field locally even if the albedo is known, since an ambiguity remains. The local ambiguity is shown to be removable in most cases by global integrability constraints. Mecca and Falcone [22] extended these works, proving a uniqueness result for weak (Lipschitz continuous) solutions in absence of shadows. In particular, the proof is based on the complete knowledge of boundary condition. This is not a reasonable assumption as this information is not directly available in the pictures of the object. They also proposed two approximation schemes for the numerical solution of this problem: an upwind finite difference scheme and a semi-Lagrangian scheme.

Tankus and Kiryati [35] changed the common orthographic projection assumption and far away light sources in Photometric Stereo to a perspective one (similar to Tankus et al. [39] in Shape from Shading). By doing that, they found an analytic linear solution for the gradient field of a 3-image perspective Photometric Stereo problem. Yoon et al. [43] employed a variational framework in their perspective Photometric Stereo algorithm, and demonstrated it using a large sets of input images (≥ 16).

Whereas orthographic PS_2 has been investigated for extracting more information from each equation using Shape from Shading techniques, and 3-image perspective Photometric

Stereo has an analytical solution for the gradient field, no information is available on the PS_2 problem under the perspective projection model. The goal of this research is to use numerical schemes commonly used in the Shape from Shading realm ([5]) also for PS_2 under the perspective projection assumption, thus extracting additional information from each given image. We write the "photometric PDEs" with the same form for both the perspective and the orthographic cases, thus "merging" the mathematical treatment of the two. We prove a uniqueness result for weak (Lipschitz continuous) solutions under the perspective projection model, and propose two numerical approximation schemes: an upwind finite difference scheme and a semi-Lagrangian scheme. Based on the unified PDEs, the convergence of the two numerical schemes can be easily deduced from [21]. This paper thus extends and combines three research directions, by Mecca and Falcone [22], Tankus and Kiryati [35], and Onn and Bruckstein [26].

In this paper, we employ the well-posed differential formulation by Mecca and Falcone [22] for the PS_3 problem with missing data (e.g., due to shadows) which adds further complexity to surface recovery. Let us emphasize that our formulation does not use geometry information due to shadows [9]. Our new approach uses the new PS_3 formulation in regions of the image where all the data is available (i.e. no shadows) and the PS_2 problem, in regions with missing parts. We thus formulate a new way to recover the shape by orienting a characteristic field in order to avoid the need in boundary condition and overpass the supplementary problem of missing part in the images. We prove that this new way to "manipulate" the characteristic field easily handles the case where images contain occlusions. A second advantage is that this new technique uses the fast-marching method, so convergence can theoretically be reached in just one iteration (in practice, the algorithm stops after very few iterations). We prove that such characteristic-field manipulation is well-posed.

Although our hypotheses are slightly weaker than the ones assumed in [15], which addressed the same problem, we consider a two step procedure and regularization terms for smooth surfaces.

The paper is organized as follows. The first part (Sect. 2) introduces the perspective in the new PS model: after the description of notations and assumptions, we formulate the new differential model for the Photometric Stereo problem (Sect. 2.1). We then prove the uniqueness of weak solution for the new differential model (Sect. 2.2). The second part (Sect. 3) is dedicated to the multi-image PS differential model: using an on/off weighted approach (Sect. 3.1), passing through an interesting remark about uniqueness of solution for this new approach (Sect. 3.2), studying the stability perturbed by noise in the images (Sect. 3.3). After recalling the semi-Lagrangian and upwind schemes used in this paper (Sect. 4), we describe how to use them for the surface recovery without taking into account boundary condition (Sect. 5). Finally, we conclude with the numerical tests on synthetic and real data (Sect. 6). Concluding remarks appear in Section 7.

2. Perspective Photometric Stereo: Notations and Assumptions. The main ingredients for the formulation of the model for the Perspective Shape from Shading (PSfS) for a Lambertian surface presented in [39] are the following:

- the light source is given by a unit vector $\omega = (\omega_1, \omega_2, \omega_3)$ (with $\omega_3 < 0$);
- the surface in the real world is given by the analytical function $h(x, y) = (x, y, \hat{z}(x, y))$

(where the point (x, y) is in the image domain $\overline{\Omega} = \Omega \cup \partial\Omega$, on the optical plane);

- the associated perspective surface is given by the function $k(\xi, \eta) = (\xi, \eta, z(\xi, \eta))$ where the point (ξ, η) is in the perspective image domain $\overline{\Omega}^p = \Omega \cup \partial\Omega$, on the focal plane (shown in blue in Fig. 2.1), parallel to the optical one (in red in Fig. 2.1) at a focal distance f;
- the transformation used to pass from one point in the optical plane (x, y) to the respective one in the focal plane is $\xi = -\frac{x}{\hat{z}(x,y)} f$, $\eta = -\frac{y}{\hat{z}(x,y)} f$. Then we have: $k(\xi, \eta) = (\xi, \eta, z(\xi, \eta)) = \big(-\frac{x}{\hat{z}(x,y)} f, -\frac{y}{\hat{z}(x,y)} f, \hat{z}(x, y)\big)$.

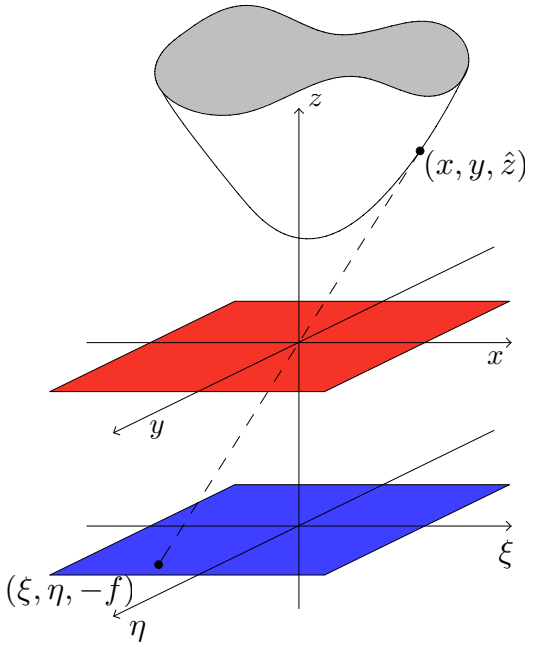


Figure 2.1. *Schematic representation of a surface taken under perspective view. The point of the real surface $(x, y, \hat{z})$ is projected in the perspective domain in the point (ξ, η) of the focal plane (in blue), parallel to the optical one (in red) at a focal distance f.*

Let us assume that the perspective surface $z(\xi, \eta)$ has no points where the view is occluded by the surface itself, this means that, once we start to consider a surface $\hat{z}(x, y)$ we have to choose a focal length $f > 0$ such that:

$$\hat{z}\left(x - t\frac{xf}{\hat{z}(x,y)}, y - t\frac{yf}{\hat{z}(x,y)}\right) < \hat{z}(x, y) + \frac{\hat{z}(x, y)}{\sqrt{x^2 + y^2}} \quad \forall t > 0, \quad \forall (x, y) \in \Omega. \tag{2.1}$$

However it is always possible to take the domain Ω small enough in order to have no occluded points.

2.1. The new Photometric Stereo Differential Model. In this section we exploit the results obtained in [22] generalising the differential model of the orthographic case to the perspective one. We consider the irradiance equation given by the inner product between the

light source ω and the normal vector to the surface $k(\xi,\eta)$. We have the following PDE in the perspective domain:

$$\rho(\xi,\eta)\frac{-fz_\xi\omega_1 - fz_\eta\omega_2 - (z+\xi z_\xi+\eta z_\eta)\omega_3}{\sqrt{f^2(z_\xi^2+z_\eta^2)+(z+\xi z_\xi+\eta z_\eta)^2}} = I(\xi,\eta), \quad \text{for} \quad (\xi,\eta)\in\Omega \tag{2.2}$$

where the unknown $\rho(\xi,\eta)$ is the albedo which represents the percentage of the light reflected by the Lambertian object. The reflectance equation (2.2) brings us to the following well known ([39]) differential problem (non-linear PDE + Dirichlet boundary condition):

$$\begin{cases} \rho(\xi,\eta)\dfrac{-z_\xi(f\omega_1+\xi\omega_3)-z_\eta(f\omega_2+\eta\omega_3)-z\omega_3}{\sqrt{f^2(z_\xi^2+z_\eta^2)+(z+\xi z_\xi+\eta z_\eta)^2}} = I(\xi,\eta), & \text{on } \Omega \\ z(\xi,\eta) = g(\xi,\eta) & \text{on } \partial\Omega \end{cases} \tag{2.3}$$

which has no unique solution (even in the viscosity solution sense).

Let us try and overcome the problem of uniqueness of solution considering the Photometric Stereo (PS) approach using two light sources defined by the unit vectors $\omega' = (\omega_1', \omega_2', \omega_3')$ and $\omega'' = (\omega_1'', \omega_2'', \omega_3'')$ (with ω_3', $\omega_3'' < 0$).

Using the information obtained by both images, we can couple the two equations related to the irradiance equation (2.2), obtaining the following system of non-linear PDEs:

$$\begin{cases} \rho(\xi,\eta)\dfrac{-z_\xi(f\omega_1'+\xi\omega_3')-z_\eta(f\omega_2'+\eta\omega_3')-z\omega_3'}{\sqrt{f^2(z_\xi^2+z_\eta^2)+(z+\xi z_\xi+\eta z_\eta)^2}} = I_1(\xi,\eta), & \text{on } \Omega \\ \rho(\xi,\eta)\dfrac{-z_\xi(f\omega_1''+\xi\omega_3'')-z_\eta(f\omega_2''+\eta\omega_3'')-z\omega_3''}{\sqrt{f^2(z_\xi^2+z_\eta^2)+(z+\xi z_\xi+\eta z_\eta)^2}} = I_2(\xi,\eta), & \text{on } \Omega \\ z(\xi,\eta) = g(\xi,\eta) & \text{on } \partial\Omega. \end{cases} \tag{2.4}$$

Now, observing that the non-linearity in the denominator and the albedo of both equations are the same (i.e. independent on the light sources) and obviously always different from zero, we can merge the non-linear equation according to the following chain of equalities:

$$\underbrace{\frac{-z_\xi(f\omega_1'+\xi\omega_3')-z_\eta(f\omega_2'+\eta\omega_3')-z\omega_3'}{I_1(\xi,\eta)} = \overbrace{\frac{\sqrt{f^2(z_\xi^2+z_\eta^2)+(z+\xi z_\xi+\eta z_\eta)^2}}{\rho(\xi,\eta)} = \frac{-z_\xi(f\omega_1''+\xi\omega_3'')-z_\eta(f\omega_2''+\eta\omega_3'')-z\omega_3''}{I_2(\xi,\eta)}}^{\text{Second Equation}}}_{\text{First Equation}} \tag{2.5}$$

Considering only the left and right hand sides of (2.5), we obtain the following linear equation

$$\begin{aligned}[(f\omega_1'+\xi\omega_3')I_2(\xi,\eta)-(f\omega_1''+\xi\omega_3'')I_1(\xi,\eta)]\frac{\partial z}{\partial\xi}(\xi,\eta) \\ +[(f\omega_2'+\eta\omega_3')I_2(\xi,\eta)-(f\omega_2''+\eta\omega_3'')I_1(\xi,\eta)]\frac{\partial z}{\partial\eta}(\xi,\eta) \\ +(\omega_3'I_2(\xi,\eta)-\omega_3''I_1(\xi,\eta))z(\xi,\eta)=0\end{aligned} \tag{2.6}$$

which leads to the following linear problem

$$\begin{cases} b(\xi,\eta)\cdot\nabla z(\xi,\eta) = s(\xi,\eta)z(\xi,\eta), & \text{on } \Omega \\ z(\xi,\eta) = g(\xi,\eta) & \text{on } \partial\Omega \end{cases} \tag{2.7}$$

where:

$$\begin{aligned} b(\xi,\eta) = ((f\omega_1' + \xi\omega_3')I_2(\xi,\eta) - (f\omega_1'' + \xi\omega_3'')I_1(\xi,\eta), \\ (f\omega_2' + \eta\omega_3')I_2(\xi,\eta) - (f\omega_2'' + \eta\omega_3'')I_1(\xi,\eta)) \end{aligned} \tag{2.8}$$

and

$$s(\xi,\eta) = \omega_3''I_1(\xi,\eta) - \omega_3'I_2(\xi,\eta). \tag{2.9}$$

Note, that the functions $b(\xi,\eta)$ and $s(\xi,\eta)$ generalize the orthographic case. Using the notation of [22], the model for the orthographic case is:

$$\begin{cases} b^{orth}(x,y)\cdot\nabla\hat{z}(x,y) = s^{orth}(x,y), & \text{on } \Omega \\ \hat{z}(x,y) = g^{orth}(x,y) & \text{on } \partial\Omega \end{cases} \tag{2.10}$$

where:

$$b^{orth}(x,y) = (\omega_1'I_2(x,y) - \omega_1''I_1(x,y), \omega_2'I_2(x,y) - \omega_2''I_1(x,y)) \tag{2.11}$$

and

$$s^{orth}(x,y) = \omega_3'I_2(\xi,\eta) - \omega_3''I_1(\xi,\eta). \tag{2.12}$$

The right hand sides of the orthographic and perspective equations differ just by sign due to the different parametrization of the light sources, since $\omega_3' > 0$, $\omega_3'' < 0$ in the perspective case. The vector field $b(\xi,\eta)$ has an additional component in the perspective case.

2.2. Uniqueness of Weak Solution for the New Differential Model. As stated in the introduction, we focus on the regularity of z to allow for weak solutions, e.g. Lipschitz continuous solutions. Following the same strategy of [22] we shall show that $b(\xi,\eta)$ guarantees the propagation of the information using the characteristic strip expansion. In order to do so we start proving the following:

Lemma 1. *If there are no points $(\xi,\eta)\in\overline{\Omega}^p$ of black shadows for the image functions (i.e. $I_1(\xi,\eta)\neq 0$ and $I_2(\xi,\eta)\neq 0$), we have that $||b(\xi,\eta)||\neq 0$.*

Proof. Let us prove this result by contradiction considering that there exists a point $(\bar{\xi},\bar{\eta})\in\overline{\Omega}^p$ such that:

$$\begin{cases} (f\omega_1' + \bar{\xi}\omega_3')I_2(\bar{\xi},\bar{\eta}) - (f\omega_1'' + \bar{\xi}\omega_3'')I_1(\bar{\xi},\bar{\eta}) = 0 \\ (f\omega_2' + \bar{\eta}\omega_3')I_2(\bar{\xi},\bar{\eta}) - (f\omega_2'' + \bar{\eta}\omega_3'')I_1(\bar{\xi},\bar{\eta}) = 0. \end{cases} \tag{2.13}$$

Since we want to consider the dependence of the image functions $I_1(\xi,\eta)$ and $I_2(\xi,\eta)$ on all the other model coefficients, we explicit these functions using the irradiance equation (2.2) obtaining the following non-linear system:

$$\begin{aligned} (f\omega_1' + \bar{\xi}\omega_3')\rho(\xi,\eta)\frac{-z_\xi(f\omega_1'' + \bar{\xi}\omega_3'') - z_\eta(f\omega_2'' + \bar{\eta}\omega_3'') - z\omega_3''}{\sqrt{f^2(z_\xi^2 + z_\eta^2) + (z + \bar{\xi}z_\xi + \bar{\eta}z_\eta)^2}} \\ - (f\omega_1'' + \bar{\xi}\omega_3'')\rho(\xi,\eta)\frac{-z_\xi(f\omega_1' + \bar{\xi}\omega_3') - z_\eta(f\omega_2' + \bar{\eta}\omega_3') - z\omega_3'}{\sqrt{f^2(z_\xi^2 + z_\eta^2) + (z + \bar{\xi}z_\xi + \bar{\eta}z_\eta)^2}} = 0, \end{aligned} \tag{2.14}$$

$$(f\omega_2' + \bar{\eta}\omega_3')\rho(\xi,\eta)\frac{-z_\xi(f\omega_1'' + \bar{\xi}\omega_3'') - z_\eta(f\omega_2'' + \bar{\eta}\omega_3'') - z\omega_3''}{\sqrt{f^2(z_\xi^2 + z_\eta^2) + (z + \bar{\xi}z_\xi + \bar{\eta}z_\eta)^2}}$$
$$- (f\omega_2'' + \bar{\eta}\omega_3'')\rho(\xi,\eta)\frac{-z_\xi(f\omega_1' + \bar{\xi}\omega_3') - z_\eta(f\omega_2' + \bar{\eta}\omega_3') - z\omega_3'}{\sqrt{f^2(z_\xi^2 + z_\eta^2) + (z + \bar{\xi}z_\xi + \bar{\eta}z_\eta)^2}} = 0. \quad (2.15)$$

Noting that the common quantity

$$\frac{\rho(\xi,\eta)}{\sqrt{f^2(z_\xi^2 + z_\eta^2) + (z + \bar{\xi}z_\xi + \bar{\eta}z_\eta)^2}} \quad (2.16)$$

does not vanish, we can remove it obtaining:

$$(f\omega_1'' + \bar{\xi}\omega_3'')[z_\xi(f\omega_1' + \bar{\xi}\omega_3') + z_\eta(f\omega_2' + \bar{\eta}\omega_3') + z\omega_3']$$
$$- (f\omega_1' + \bar{\xi}\omega_3')[z_\xi(f\omega_1'' + \bar{\xi}\omega_3'') + z_\eta(f\omega_2'' + \bar{\eta}\omega_3'') + z\omega_3''] = 0, \quad (2.17)$$

$$(f\omega_2'' + \bar{\eta}\omega_3'')[z_\xi(f\omega_1' + \bar{\xi}\omega_3') + z_\eta(f\omega_2' + \bar{\eta}\omega_3') + z\omega_3']$$
$$- (f\omega_2' + \bar{\eta}\omega_3')[z_\xi(f\omega_1'' + \bar{\xi}\omega_3'') + z_\eta(f\omega_2'' + \bar{\eta}\omega_3'') + z\omega_3''] = 0. \quad (2.18)$$

This system can be written as:

$$\bar{\xi}(\omega_2'\omega_3'' - \omega_2''\omega_3')z_\eta + \bar{\eta}(\omega_1''\omega_3' - \omega_1'\omega_3'')z_\eta = (\omega_1'\omega_3'' - \omega_1''\omega_3')z + f(\omega_1'\omega_2'' - \omega_1''\omega_2')z_\eta, \quad (2.19)$$
$$\bar{\xi}(\omega_2'\omega_3'' - \omega_2''\omega_3')z_\xi + \bar{\eta}(\omega_1''\omega_3' - \omega_1'\omega_3'')z_\xi = (\omega_2''\omega_3' - \omega_2'\omega_3'')z + f(\omega_1'\omega_2'' - \omega_1''\omega_2')z_\xi. \quad (2.20)$$

and it gives an explicit formula for the first partial derivative as follows

$$z_\eta = \frac{(\omega_1'\omega_3'' - \omega_1''\omega_3')z}{\bar{\xi}(\omega_2'\omega_3'' - \omega_2''\omega_3') + \bar{\eta}(\omega_1''\omega_3' - \omega_1'\omega_3'') - f(\omega_1'\omega_2'' - \omega_1''\omega_2')} \quad (2.21)$$
$$z_\xi = \frac{(\omega_2''\omega_3' - \omega_2'\omega_3'')z}{\bar{\xi}(\omega_2'\omega_3'' - \omega_2''\omega_3') + \bar{\eta}(\omega_1''\omega_3' - \omega_1'\omega_3'') - f(\omega_1'\omega_2'' - \omega_1''\omega_2')}. \quad (2.22)$$

Replacing (2.21) and (2.22) in the irradiance equations of (2.4) we get $I_1 = I2 = 0$, which is in contradiction with the assumption of absence of shadows. Let us consider the other particular cases as follows:

$$case\ A \equiv \begin{cases} \omega_2'\omega_3'' - \omega_2''\omega_3' = 0 \\ \omega_1''\omega_3' - \omega_1'\omega_3'' = 0 \\ \omega_1'\omega_2'' - \omega_1''\omega_2' = 0 \end{cases}, \quad case\ B \equiv \begin{cases} \omega_2'\omega_3'' - \omega_2''\omega_3' = 0 \\ \omega_1''\omega_3' - \omega_1'\omega_3'' = 0 \\ \omega_1'\omega_2'' - \omega_1''\omega_2' \neq 0 \\ z_\xi = z_\eta = 0 \end{cases}. \quad (2.23)$$

From a geometric point of view, the configurations leading to these equations are impossible. Let us consider the light vectors as generators of the light planes, namely π' and π'' which are orthogonal to the respective vectors ω' and ω'' (see Fig. 2.2):

$$\begin{aligned} \pi' &: \quad \omega_1'x + \omega_2'y + \omega_3'z = 0 \\ \pi'' &: \quad \omega_1''x + \omega_2''y + \omega_3''z = 0. \end{aligned} \quad (2.24)$$

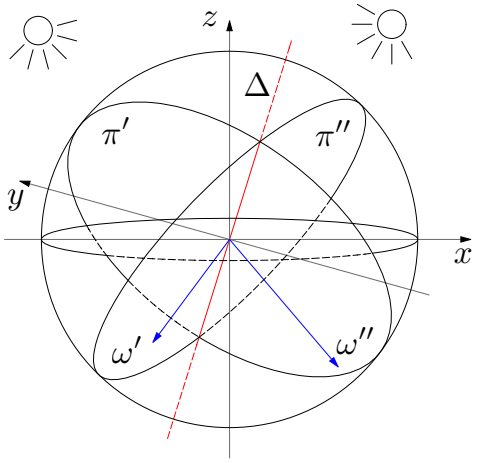


Figure 2.2. *The connection between the degenerate case of the linear system made by equations (2.19) and (2.20) and the line (Δ) of intersection between the two light planes.*

Once we defined the planes π' and π'' we can easily compute the vector of the straight line Δ generated by the intersection of these two planes simply by the cross product between ω' and ω'':

$$\omega' \times \omega'' = \begin{pmatrix} \omega'_2\omega''_3 - \omega''_2\omega'_3 \\ \omega''_1\omega'_3 - \omega'_1\omega''_3 \\ \omega'_1\omega''_2 - \omega''_1\omega'_2 \end{pmatrix} = \begin{pmatrix} l \\ m \\ n \end{pmatrix}. \tag{2.25}$$

Now, the only possibility to verify *case A* is to choose $\omega' \equiv \omega''$ which does not respect the main PS assumption. If we consider *case B*, it is clear that the condition $l = 0$, $m = 0$ and $n \neq 0$ (i.e. the straight line Δ is vertical) can be reached only if $\omega'_3 = \omega''_3 = 0$ which contradicts the assumption: $\omega'_3, \omega''_3 < 0$.

■

We will see that this lemma will be useful in two important points in the sequel: the first is for the proof of the uniqueness of solution of the problem, the second is related to an important remark about the semi-Lagrangian numerical scheme.

In order to complete the theoretical analysis we will extend the uniqueness results of the differential problem (2.7) for the case of a weak solution as in the orthographic case. Discussion of depth-discontinuities and multiple objects is beyond the scope of this paper. Our purpose is to prove the uniqueness of solution of (2.7) in the Lipschitz function space via characteristics method. The meaning of weak solution here is intended as a combination of classical solutions, each defined on a different domain. These domains are then going to be patched together in such a way that, across the boundaries between domains on which there are discontinuities in some derivatives, the equation (2.7) is satisfied. Let us recall that the points where the surface z is nondifferentiable are the same where the functions b and s are discontinuous (jump discontinuity) [22]. We assume the points of discontinuity are given as a family of regular curves $(\gamma_1(t), \ldots, \gamma_k(t))$ where t is the argument of the parametric representation. It is clear that this curve contains also the points of discontinuity of the image functions $I_1(\xi, \eta)$

and $I_2(\xi,\eta)$ (see Fig. 2.3 of Example 1 and Fig. 2.4 of Example 2). Now, since the functions $b(\xi,\eta)$ and $s(\xi,\eta)$ depend directly on $I_1(\xi,\eta)$ and $I_2(\xi,\eta)$, the same family of curves represents the discontinuity also for these coefficients of the PDE in (2.7).

Another property of the discontinuity type of $b(\xi,\eta)$ and $s(\xi,\eta)$ is due to the fact that we look for a Lipschitz solution, so the discontinuity must be a jump discontinuity.

Example 1. We show here with an example how the point of discontinuity in the image functions can be interpreted as points where the surface is nondifferentiable. Let us simplify the geometry of the example considering the following function in dimension one:

$$\hat{z}(x) = \begin{cases} 3 - x & x \geq 0 \\ 3 + 2x & x < 0 \end{cases} \tag{2.26}$$

defined in the domain $\Omega = [a,b] = [-1,1]$. Once we fix the focal length, for example $f = 3$ (see Fig. 2.3), we can easily compute the perspective domain

$$\Omega = [a^p, b^p] = \left[-\frac{b}{\hat{z}(b)} f, -\frac{a}{\hat{z}(a)} f \right] = \left[-\frac{3}{2}, 3 \right]. \tag{2.27}$$

In this particular case, the perspective function can be computed accurately. Exploiting the knowledge of $\hat{z}$ and that $\hat{z}(0) = z(0) = 3$ we can compute $z(\xi)$ since it consists of a couple of straight lines (see Fig. 2.3). In fact:

$$z(\xi) = \begin{cases} r_1 \text{ straight line passing through } (-\frac{3}{2}, 2) \text{ and } (0,3) \\ r_2 \text{ straight line passing through } (0,3) \text{ and } (3,1) \end{cases} \tag{2.28}$$

that is

$$z(\xi) = -\frac{2}{3}|\xi| + 3, \quad \xi \in \left[-\frac{3}{2}, 3 \right]. \tag{2.29}$$

If we consider the inner product between the normal field to this function with a non-vertical light vector $\omega = (\omega_1, \omega_2)$ it is clear that the image function has a jump discontinuity at the same point where the function z is nondifferentiable. We highlight the coincidence for which the discontinuity of the image function and the non-differentiability of $\hat{z}$ are at the same point (see Fig. 2.3). This depends only on the fact that this point remains fixed with respect the perspective transformation.

Example 2. Let us consider now another example in order to give a better explanation about the position of the points of nondifferentiability. Let us examine a more complex function defined in $\Omega = [-2,4]$ (see Fig. 2.4) as follows:

$$\hat{z}(x) = \begin{cases} |x+1| + 6 & x \in [-2,1] \\ -2x + 10 & x \in [1,3] \\ x + 1 & x \in [3,4]. \end{cases} \tag{2.30}$$

Let us list the coordinates on the x-axis where it is nondifferentiable in this order: $p_1 = -1$, $p_2 = 1$ and $p_3 = 3$. With respect to this function, we have that the perspective domain

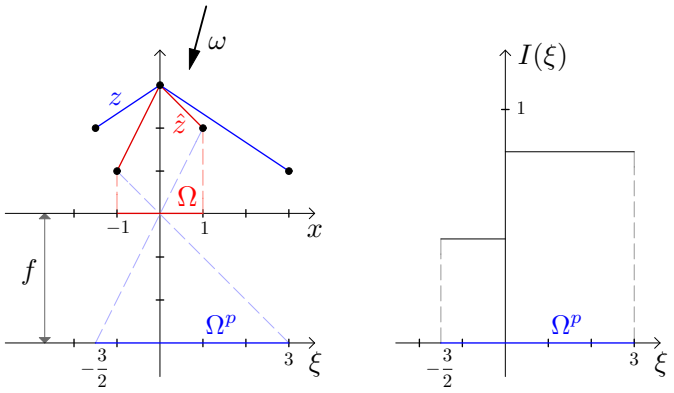


Figure 2.3. *A schematic example shows that the points where the image function $I(\xi)$ is discontinuous (in the focal plane) are in one-to-one correspondence with the points of the optical plane for which the solution $\hat{z}(x)$ is nondifferentiable. On the left, the Lipschitz "surfaces" in the real world $\hat{z}(x)$ (in red) and the associated perspective transformation z (in blue). We emphasize also the two domains Ω and Ω using the same respective colors. On the right is the associated image through the light ω.*

(considering a focal length $f = 3$) restricts to $\Omega = \left[-\frac{12}{5}, \frac{6}{7}\right]$. The associated perspective function can be explicitly written as follows:

$$z(\xi) = \begin{cases} -\frac{20}{3}\xi - 16 & \xi \in [-\frac{12}{5}, -\frac{9}{4}] \\ \frac{32}{15}\xi + \frac{132}{15} & \xi \in [-\frac{9}{4}, -\frac{3}{8}] \\ -\frac{16}{7}\xi + \frac{50}{7} & \xi \in [-\frac{3}{8}, \frac{1}{2}] \\ \frac{14}{5}\xi + \frac{23}{5} & \xi \in [\frac{1}{2}, \frac{6}{7}] \end{cases} \tag{2.31}$$

(see Fig. 2.4).

Looking at the image function $I(\xi)$ (obtained with light source direction ω, both shown in Fig. 2.4), we can see that the point of discontinuity are shifted with respect to the points where the function is nondifferentiable in the real world. In fact, according to the perspective transformation the points where the surface $z(\xi)$ is nondifferentiable are now shifted as follows:

$$\begin{array}{lll} p_1 & \to & p_1^p = -\frac{9}{4} \\ p_2 & \to & p_2^p = -\frac{3}{8} \\ p_3 & \to & p_3^p = \frac{1}{2}. \end{array} \tag{2.32}$$

Theorem 2.

Let $\gamma(t)$ be a curve of discontinuity for the function $b(\xi, \eta)$ (and $s(\xi, \eta)$) and let $\overline{p} = (\overline{\xi}, \overline{\eta})$ be a point of this curve. Let $n(\overline{\xi}, \overline{\eta})$ be the outgoing normal with respect to the set Ω_+^p, then we have

$$\left[\lim_{\substack{(\xi,\eta)\to(\overline{\xi},\overline{\eta}) \\ (\xi,\eta)\in\Omega_+}} b(\xi,\eta)\cdot n(\overline{\xi},\overline{\eta})\right]\left[\lim_{\substack{(\xi,\eta)\to(\overline{\xi},\overline{\eta}) \\ (\xi,\eta)\in\Omega_-}} b(\xi,\eta)\cdot n(\overline{\xi},\overline{\eta})\right] \geq 0 \tag{2.33}$$

Proof.

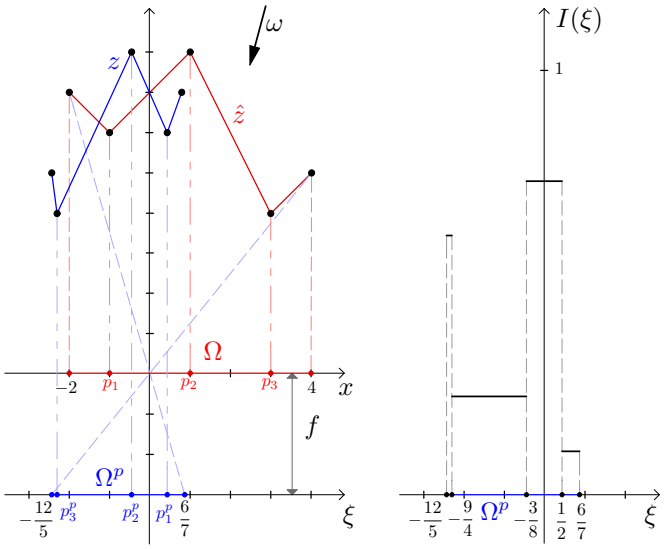


Figure 2.4. *On the left the real (red) end the perspective (blue) Lipschitz functions used to explain how the nondifferentiable points for $\hat{z}$ shift according to the perspective transformation z. It is also possible to note the different domains in the respective colors. On the right the image $I(\xi)$ associated to the perspective function z obtained with the light source ω represented on the left picture.*

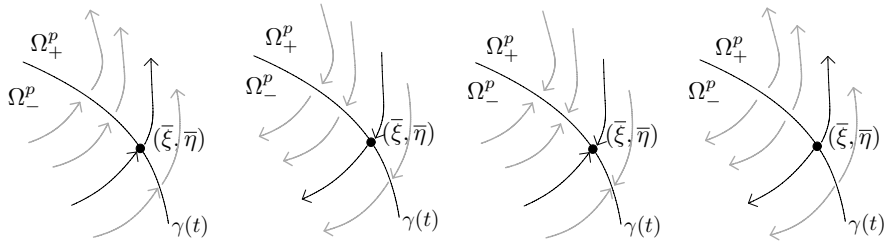


Figure 2.5. *All the possible behaviors of the characteristic field close to the discontinuity curve $\gamma(t)$. The only admissible cases (that permit information to travel along the characteristic curves) are the first two (from the left).*

Let us start by defining the following quantities:

$$I_1^+ = \lim_{\substack{(\xi,\eta)\to(\bar{\xi},\bar{\eta}) \\ (\xi,\eta)\in\Omega_+}} I_1(\xi,\eta) = \lim_{\substack{(\xi,\eta)\to(\bar{\xi},\bar{\eta}) \\ (\xi,\eta)\in\Omega_+}} \rho(\xi,\eta)\frac{-\nabla z(\xi,\eta)\cdot(f\widetilde{\omega}' + (\xi,\eta)\omega_3') - z(\xi,\eta)\omega_3'}{\sqrt{f|\nabla z(\xi,\eta)|^2 + (z(\xi,\eta) + (\xi,\eta)\cdot\nabla z(\xi,\eta))^2}} =$$

$$\rho^+\frac{-\nabla z^+\cdot(f\widetilde{\omega}' + \overline{p}\omega_3') - \overline{z}\omega_3'}{\sqrt{f|\nabla z^+|^2 + (\overline{z} + \overline{p}\cdot\nabla z^+)^2}},$$

$$I_1^- = \lim_{\substack{(\xi,\eta)\to(\overline{\xi},\overline{\eta})\\(\xi,\eta)\in\Omega_-}} I_1(\xi,\eta) = \lim_{\substack{(\xi,\eta)\to(\overline{\xi},\overline{\eta})\\(\xi,\eta)\in\Omega_-}} \rho(\xi,\eta)\frac{-\nabla z(\xi,\eta)\cdot(f\widetilde{\omega}' + (\xi,\eta)\omega_3') - z(\xi,\eta)\omega_3'}{\sqrt{f|\nabla z(\xi,\eta)|^2 + (z(\xi,\eta)+(\xi,\eta)\cdot\nabla z(\xi,\eta))^2}} =$$

$$\rho^- \frac{-\nabla z^-\cdot(f\widetilde{\omega}' + \overline{p}\omega_3') - \overline{z}\omega_3'}{\sqrt{f|\nabla z^-|^2 + (\overline{z}+\overline{p}\cdot\nabla z^-)^2}}$$

$$I_2^+ = \lim_{\substack{(\xi,\eta)\to(\overline{\xi},\overline{\eta})\\(\xi,\eta)\in\Omega_+}} I_2(\xi,\eta) = \lim_{\substack{(\xi,\eta)\to(\overline{\xi},\overline{\eta})\\(\xi,\eta)\in\Omega_+}} \rho(\xi,\eta)\frac{-\nabla z(\xi,\eta)\cdot(f\widetilde{\omega}'' + (\xi,\eta)\omega_3'') - z(\xi,\eta)\omega_3''}{\sqrt{f|\nabla z(\xi,\eta)|^2 + (z(\xi,\eta)+(\xi,\eta)\cdot\nabla z(\xi,\eta))^2}} =$$

$$\rho^+ \frac{-\nabla z^+\cdot(f\widetilde{\omega}'' + \overline{p}\omega_3'') - \overline{z}\omega_3''}{\sqrt{f|\nabla z^+|^2 + (\overline{z}+\overline{p}\cdot\nabla z^+)^2}},$$

$$I_2^- = \lim_{\substack{(\xi,\eta)\to(\overline{\xi},\overline{\eta})\\(\xi,\eta)\in\Omega_-}} I_2(\xi,\eta) = \lim_{\substack{(\xi,\eta)\to(\overline{\xi},\overline{\eta})\\(\xi,\eta)\in\Omega_-}} \rho(\xi,\eta)\frac{-\nabla z(\xi,\eta)\cdot(f\widetilde{\omega}'' + (\xi,\eta)\omega_3'') - z(\xi,\eta)\omega_3''}{\sqrt{f|\nabla z(\xi,\eta)|^2 + (z(\xi,\eta)+(\xi,\eta)\cdot\nabla z(\xi,\eta))^2}} =$$

$$\rho^- \frac{-\nabla z^-\cdot(f\widetilde{\omega}'' + \overline{p}\omega_3'') - \overline{z}\omega_3''}{\sqrt{f|\nabla z^-|^2 + (\overline{z}+\overline{p}\cdot\nabla z^-)^2}}$$

where $\overline{z} = z(\overline{\xi},\overline{\eta})$ is a well defined quantity since we assume the surface z is Lipschitz continuous and

$$\rho^+ = \lim_{\substack{(\xi,\eta)\to(\overline{\xi},\overline{\eta})\\(\xi,\eta)\in\Omega_+}} \rho(\xi,\eta), \quad \rho^- = \lim_{\substack{(\xi,\eta)\to(\overline{\xi},\overline{\eta})\\(\xi,\eta)\in\Omega_-}} \rho(\xi,\eta)$$
$$\nabla z^+ = \lim_{\substack{(\xi,\eta)\to(\overline{\xi},\overline{\eta})\\(\xi,\eta)\in\Omega_+}} \nabla z(\xi,\eta), \quad \nabla z^- = \lim_{\substack{(\xi,\eta)\to(\overline{\xi},\overline{\eta})\\(\xi,\eta)\in\Omega_-}} \nabla z(\xi,\eta).$$

Now, we replace the previous limit values for the image functions in the limit definition for b, we have:

$$(b_1^+, b_2^+) = \lim_{\substack{(\xi,\eta)\to(\overline{\xi},\overline{\eta})\\(\xi,\eta)\in\Omega_+}} b(\xi,\eta) =$$
$$((f\omega_1' + \overline{\xi}\omega_3')I_2^+ - (f\omega_1'' + \overline{\xi}\omega_3'')I_1^+, (f\omega_2' + \overline{\eta}\omega_3')I_2^+ - (f\omega_2'' + \overline{\eta}\omega_3'')I_1^+) \quad (2.34)$$

$$(b_1^-, b_2^-) = \lim_{\substack{(\xi,\eta)\to(\overline{\xi},\overline{\eta})\\(\xi,\eta)\in\Omega_-}} b(\xi,\eta) =$$
$$((f\omega_1' + \overline{\xi}\omega_3')I_2^- - (f\omega_1'' + \overline{\xi}\omega_3'')I_1^-, (f\omega_2' + \overline{\eta}\omega_3')I_2^- - (f\omega_2'' + \overline{\eta}\omega_3'')I_1^-) \quad (2.35)$$

If we call the two components of the normal vector $n(\overline{\xi},\overline{\eta})$: n_1 and n_2 (both fixed but depend on $(\overline{\xi},\overline{\eta})$), we can write (2.33) as follows:

$$\begin{aligned}(b_1^+ n_1 + b_2^+ n_2)(b_1^- n_1 + b_2^- n_2) \geq 0,\\ b_1^+ b_1^- n_1^2 + b_2^+ b_2^- n_2^2 + (b_1^+ b_2^- + b_1^- b_2^+) n_1 n_2 \geq 0.\end{aligned} \quad (2.36)$$

Using (2.34) and (2.35) we have:

$$\begin{aligned}&\big((f\omega_1'+\overline{\xi}\omega_3')I_2^+-(f\omega_1''+\overline{\xi}\omega_3'')I_1^+\big)\big((f\omega_1'+\overline{\xi}\omega_3')I_2^--(f\omega_1''+\overline{\xi}\omega_3'')I_1^-\big)n_1^2+\\&\big((f\omega_2'+\overline{\eta}\omega_3')I_2^+-(f\omega_2''+\overline{\eta}\omega_3'')I_1^+\big)\big((f\omega_2'+\overline{\eta}\omega_3')I_2^--(f\omega_2''+\overline{\eta}\omega_3'')I_1^-\big)n_2^2+\\&\big[\big((f\omega_1'+\overline{\xi}\omega_3')I_2^+-(f\omega_1''+\overline{\xi}\omega_3'')I_1^+\big)\big((f\omega_2'+\overline{\eta}\omega_3')I_2^--(f\omega_2''+\overline{\eta}\omega_3'')I_1^-\big)+\\&\big((f\omega_1'+\overline{\xi}\omega_3')I_2^--(f\omega_1''+\overline{\xi}\omega_3'')I_1^-\big)\big((f\omega_2'+\overline{\eta}\omega_3')I_2^+-(f\omega_2''+\overline{\eta}\omega_3'')I_1^+\big)\big]n_1n_2\geq 0\end{aligned}\tag{2.37}$$

and splitting the computation into three parts with respect to the coefficients of n_1^2, n_2^2 and n_1n_2 we have the following three results, respectively:

$$\begin{aligned}&I_1^+I_1^-\big(f\omega_1''+\omega_3''\overline{\xi}\big)^2+I_2^+I_2^-\big(f\omega_1'+\omega_3'\overline{\xi}\big)^2-\\&\qquad(I_1^+I_2^-+I_1^-I_2^+)\big(f^2\omega_1'\omega_1''+f\omega_1''\omega_3'\overline{\xi}+f\omega_1'\omega_3''\overline{\xi}+\omega_3'\omega_3''\overline{\xi}^2\big)=\\&\qquad I_1^+I_1^-\big(f^2(\omega_1'')^2+2f\omega_1''\omega_3''\overline{\xi}+(\omega_3'')^2\overline{\xi}^2\big)+I_2^+I_2^-\big(f^2(\omega_1')^2+2f\omega_1'\omega_3'\overline{\xi}+(\omega_3')^2\overline{\xi}^2\big)-\\&I_1^+I_2^-\big(f^2\omega_1'\omega_1''+f\omega_1''\omega_3'\overline{\xi}+f\omega_1'\omega_3''\overline{\xi}+\omega_3'\omega_3''\overline{\xi}^2\big)-I_1^-I_2^+\big(f^2\omega_1'\omega_1''+f\omega_1''\omega_3'\overline{\xi}+f\omega_1'\omega_3''\overline{\xi}+\omega_3'\omega_3''\overline{\xi}^2\big),\end{aligned}\tag{2.38}$$

$$\begin{aligned}&I_1^+I_1^-\big(f\omega_2''+\omega_3''\overline{\eta}\big)^2+I_2^+I_2^-\big(f\omega_2'+\omega_3'\overline{\eta}\big)^2-\\&\qquad(I_1^+I_2^-+I_1^-I_2^+)\big(f^2\omega_2'\omega_2''+f\omega_2''\omega_3'\overline{\eta}+f\omega_2'\omega_3''\overline{\eta}+\omega_3'\omega_3''\overline{\eta}^2\big)=\\&\qquad I_1^+I_1^-\big(f^2(\omega_2'')^2+2f\omega_2''\omega_3''\overline{\eta}+(\omega_3'')^2\overline{\eta}^2\big)+I_2^+I_2^-\big(f^2(\omega_2')^2+2f\omega_2'\omega_3'\overline{\eta}+(\omega_3')^2\overline{\eta}^2\big)-\\&I_1^+I_2^-\big(f^2\omega_1'\omega_2''+f\omega_2''\omega_3'\overline{\eta}+f\omega_2'\omega_3''\overline{\eta}+\omega_3'\omega_3''\overline{\eta}^2\big)-I_1^-I_2^+\big(f^2\omega_2'\omega_1''+f\omega_2''\omega_3'\overline{\eta}+f\omega_2'\omega_3''\overline{\eta}+\omega_3'\omega_3''\overline{\eta}^2\big),\end{aligned}\tag{2.39}$$

$$\begin{aligned}&2I_1^+I_1^-\big(f^2\omega_1''\omega_2''+f\omega_1''\omega_3''\overline{\eta}+f\omega_2''\omega_3''\overline{\xi}+(\omega_3'')^2\overline{\xi}\overline{\eta}\big)+\\&\qquad2I_2^+I_2^-\big(f^2\omega_1'\omega_2'+f\omega_1'\omega_3'\overline{\eta}+f\omega_2'\omega_3'\overline{\xi}+(\omega_3')^2\overline{\xi}\overline{\eta}\big)-\\&(I_1^+I_2^-+I_1^-I_2^+)\big(f^2\omega_1''\omega_2'+f^2\omega_1'\omega_2''+f\omega_1''\omega_3'\overline{\eta}+f\omega_1'\omega_3''\overline{\eta}+f\omega_2''\omega_3'\overline{\xi}+f\omega_2'\omega_3''\overline{\xi}+2\omega_3'\omega_3''\overline{\xi}\overline{\eta}\big).\end{aligned}\tag{2.40}$$

The image functions $I_1^+I_1^-$, $I_1^+I_2^-$, $I_1^-I_2^+$ and $I_2^+I_2^-$ appearing in (2.38), (2.39) and (2.40) have the common quantity

$$\frac{\rho^+\rho^-}{\sqrt{f^2|\nabla z^+|^2+(\overline{z}+\overline{p}\cdot\nabla z^+)^2}\sqrt{f^2|\nabla z^-|^2+(\overline{z}+\overline{p}\cdot\nabla z^-)^2}}\tag{2.41}$$

that can be eliminated since it is always strictly positive. Only the following quantities can be then considered:

$$\begin{aligned}i_1^+&=-\nabla z^+\cdot(f\widetilde{\omega}'+(\overline{\xi},\overline{\eta})\omega_3')-\overline{z}\omega_3',\quad i_1^-=-\nabla z^-\cdot(f\widetilde{\omega}'+(\overline{\xi},\overline{\eta})\omega_3')-\overline{z}\omega_3',\\i_2^+&=-\nabla z^+\cdot(f\widetilde{\omega}''+(\overline{\xi},\overline{\eta})\omega_3'')-\overline{z}\omega_3'',\quad i_2^-=-\nabla z^-\cdot(f\widetilde{\omega}''+(\overline{\xi},\overline{\eta})\omega_3'')-\overline{z}\omega_3'',\end{aligned}\tag{2.42}$$

which allow us to write the grayscale value of one image with respect to the grayscale value of the same image but taken on the other side, that is:

$$i_1^+=i_1^-+\lambda\cdot(f\widetilde{\omega}'+(\overline{\xi},\overline{\eta})\omega_3'),\qquad i_2^+=i_2^-+\lambda\cdot(f\widetilde{\omega}''+(\overline{\xi},\overline{\eta})\omega_3'')\tag{2.43}$$

where

$$\lambda = (\nabla z^- - \nabla z^+) = (\lambda_1, \lambda_2) = \left(\frac{\partial z^-}{\partial x} - \frac{\partial z^+}{\partial x}, \frac{\partial z^-}{\partial y} - \frac{\partial z^+}{\partial y} \right). \tag{2.44}$$

Now, we substitute these quantities in (2.38), (2.39) and (2.40) obtaining the following three coefficients for n_1^2, n_2^2 and $n_1 n_2$:

$$\begin{aligned}
&(i_1^- + \lambda \cdot (f\widetilde{\omega}' + (\overline{\xi}, \overline{\eta})\omega_3'))i_1^- \left(f^2(\omega_1'')^2 + 2f\omega_1''\omega_3''\overline{\xi} + (\omega_3'')^2\overline{\xi}^2 \right) + \\
&\qquad (i_2^- + \lambda \cdot (f\widetilde{\omega}'' + (\overline{\xi}, \overline{\eta})\omega_3''))i_2^- \left(f^2(\omega_1')^2 + 2f\omega_1'\omega_3'\overline{\xi} + (\omega_3')^2\overline{\xi}^2 \right) - \\
&\qquad (i_1^- + \lambda \cdot (f\widetilde{\omega}' + (\overline{\xi}, \overline{\eta})\omega_3'))i_2^- \left(f^2\omega_1'\omega_1'' + f\omega_1''\omega_3'\overline{\xi} + f\omega_1'\omega_3''\overline{\xi} + \omega_3'\omega_3''\overline{\xi}^2 \right) - \\
&\qquad i_1^-(i_2^- + \lambda \cdot (f\widetilde{\omega}'' + (\overline{\xi}, \overline{\eta})\omega_3''))\left(f^2\omega_1'\omega_1'' + f\omega_1''\omega_3'\overline{\xi} + f\omega_1'\omega_3''\overline{\xi} + \omega_3'\omega_3''\overline{\xi}^2 \right),
\end{aligned} \tag{2.45}$$

$$\begin{aligned}
&(i_1^- + \lambda \cdot (f\widetilde{\omega}' + (\overline{\xi}, \overline{\eta})\omega_3'))i_1^- \left(f^2(\omega_2'')^2 + 2f\omega_2''\omega_3''\overline{\eta} + (\omega_3'')^2\overline{\eta}^2 \right) + \\
&\qquad (i_2^- + \lambda \cdot (f\widetilde{\omega}'' + (\overline{\xi}, \overline{\eta})\omega_3''))i_2^- \left(f^2(\omega_2')^2 + 2f\omega_2'\omega_3'\overline{\eta} + (\omega_3')^2\overline{\eta}^2 \right) - \\
&\qquad (i_1^- + \lambda \cdot (f\widetilde{\omega}' + (\overline{\xi}, \overline{\eta})\omega_3'))i_2^- \left(f^2\omega_1'\omega_2'' + f\omega_2''\omega_3'\overline{\eta} + f\omega_2'\omega_3''\overline{\eta} + \omega_3'\omega_3''\overline{\eta}^2 \right) - \\
&\qquad i_1^-(i_2^- + \lambda \cdot (f\widetilde{\omega}'' + (\overline{\xi}, \overline{\eta})\omega_3''))\left(f^2\omega_2'\omega_1'' + f\omega_2''\omega_3'\overline{\eta} + f\omega_2'\omega_3''\overline{\eta} + \omega_3'\omega_3''\overline{\eta}^2 \right),
\end{aligned} \tag{2.46}$$

$$\begin{aligned}
&2(i_1^- + \lambda \cdot (f\widetilde{\omega}' + (\overline{\xi}, \overline{\eta})\omega_3'))i_1^- \left(f^2\omega_1''\omega_2'' + f\omega_1''\omega_3''\overline{\eta} + f\omega_2''\omega_3''\overline{\xi} + (\omega_3'')^2\overline{\xi}\overline{\eta} \right) + \\
&\qquad 2(i_2^- + \lambda \cdot (f\widetilde{\omega}'' + (\overline{\xi}, \overline{\eta})\omega_3''))i_2^- \left(f^2\omega_1'\omega_2' + f\omega_1'\omega_3'\overline{\eta} + f\omega_2'\omega_3'\overline{\xi} + (\omega_3')^2\overline{\xi}\overline{\eta} \right) - \\
&\qquad [(i_1^- + \lambda \cdot (f\widetilde{\omega}' + (\overline{\xi}, \overline{\eta})\omega_3'))i_2^- + i_1^-(i_2^- + \lambda \cdot (f\widetilde{\omega}'' + (\overline{\xi}, \overline{\eta})\omega_3''))] \\
&\qquad \left(f^2\omega_1''\omega_2' + f^2\omega_1'\omega_2'' + f\omega_1''\omega_3'\overline{\eta} + f\omega_1'\omega_3''\overline{\eta} + f\omega_2''\omega_3'\overline{\xi} + f\omega_2'\omega_3''\overline{\xi} + 2\omega_3'\omega_3''\overline{\xi}\overline{\eta} \right).
\end{aligned} \tag{2.47}$$

After some computation we have:

$$\begin{aligned}
&(2.45)n_1^2 + (2.46)n_2^2 + (2.47)n_1 n_2 = \\
&\left[f\left(i_2^-(n_1\omega_1' + n_2\omega_2') - i_1^-(n_1\omega_1'' + n_2\omega_2'') \right) + (i_2^-\omega_3' - i_1^-\omega_3'')(n_1\xi + n_2\eta) \right] \\
&\Big[- f^2(n_1\lambda_2 - n_2\lambda_1)(\omega_1''\omega_2' - \omega_1'\omega_2'') + (i_2^-\omega_3' - i1^-\omega_3'')(n_1\xi + n_2\eta) + \\
&\quad f\left(i_2^-(n_1\omega_1' + n_2\omega_2') - i_2^-(n_1\omega_1'' + n_2\omega_2'') - \right. \\
&\quad \left. (n_1\lambda_2 - n_2\lambda_1)(\eta\omega_1''\omega_3' - \eta\omega_1'\omega_3'' - \xi\omega_2''\omega_3' + \xi\omega_2'\omega_3'') \right) \Big] \overset{*}{=} \\
&\left[f\left(i_2^-(n_1\omega_1' + n_2\omega_2') - i_1^-(n_1\omega_1'' + n_2\omega_2'') \right) + (i_2^-\omega_3' - i_1^-\omega_3'')(n_1\xi + n_2\eta) \right]^2 \geq 0
\end{aligned} \tag{2.48}$$

where in * we are using that

$$(n_1\lambda_2 - n_2\lambda_1) \equiv 0 \tag{2.49}$$

exploiting the continuity of the surface z. In fact, using (2.44), we can write the previous equivalence as follows:

$$\begin{aligned}
n_1\lambda_2 - n_2\lambda_1 &= n_1\left(\frac{\partial z^-}{\partial \eta} - \frac{\partial z^+}{\partial \eta} \right) - n_2\left(\frac{\partial z^-}{\partial \xi} - \frac{\partial z^+}{\partial \xi} \right) = \\
&\quad - \frac{\partial z^-}{\partial \xi} n_2 + \frac{\partial z^-}{\partial \eta} n_1 - \left(- \frac{\partial z^+}{\partial \xi} n_2 + \frac{\partial z^+}{\partial \eta} n_1 \right) = \\
\lim_{\substack{(\xi,\eta)\to(\overline{\xi},\overline{\eta}) \\ (\xi,\eta)\in\Omega_-}} \nabla_{(-n_2,n_1)} z(\xi,\eta) &- \lim_{\substack{(\xi,\eta)\to(\overline{\xi},\overline{\eta}) \\ (\xi,\eta)\in\Omega_+}} \nabla_{(-n_2,n_1)} z(\xi,\eta) = \frac{\partial z^-}{\partial t}(\overline{\xi},\overline{\eta}) - \frac{\partial z^+}{\partial t}(\overline{\xi},\overline{\eta}) = 0
\end{aligned} \tag{2.50}$$

where in the last passage we have used the orthogonality of $(-n_2, n_1)$ and the normal vector (n_1, n_2) of the discontinuity curve $\gamma(t)$. This means that we shall consider the gradient in the tangential direction with respect to $\gamma(t)$ for both sides Ω_+ and Ω_-, that is $\frac{\partial z^+}{\partial t}$ and $\frac{\partial z^-}{\partial t}$. Now, since we are assuming continuity for surface z, these two derivatives are equal for all the points $(\overline{\xi}, \overline{\eta})$.

■

The previous theorem explains the behavior of $b(\xi, \eta)$ near a curve of discontinuity γ. Summarizing, it says that this curve is not a barrier for the vector field. We will use this theorem in the next parts for the characteristic strip expansion method. We show that, thanks to this last result, the information can travel the complete domain without obstacle.

In order to simplify the analysis of the numerical schemes we use in this paper, we modify the differential problem (2.7) dividing by $z(\xi, \eta)$ the differential equation and considering the bijective mapping $\ln(z(\xi, \eta)) = u(\xi, \eta)$. We finally get the following problem:

$$\begin{cases} b(\xi, \eta) \cdot \nabla u(\xi, \eta) = s(\xi, \eta), & \text{on } \Omega \\ u(\xi, \eta) = e^{g(\xi, \eta)} & \text{on } \partial\Omega \end{cases} \tag{2.51}$$

The substitution makes the differential problem equivalent to the orthographic problem [22].

Theorem 3.

Let us consider the problem

$$\begin{cases} b(\xi, \eta) \cdot \nabla u(\xi, \eta) = s(\xi, \eta), & a.e.\ (\xi, \eta) \in \Omega \\ u(\xi, \eta) = e^{g(\xi, \eta)} & \forall (\xi, \eta) \in \partial\Omega \end{cases} \tag{2.52}$$

where the functions $b(\xi, \eta)$ and $s(\xi, \eta)$ are defined in (2.8) and (2.9) respectively.

Let us suppose that $(\gamma_1(t), \ldots, \gamma_k(t))$, the family of discontinuity curves for $b(\xi, \eta)$ and $s(\xi, \eta)$, are not characteristic curves (with respect to the previous problem). Then there exists a unique Lipschitz solution of the problem.

We omit the proof because Lemma 1 and Theorem 2 allow to build the unique solution by following the same logical passages of the orthographic case [22], that is proving that the ODE system

$$\begin{cases} (\mathrm{a}) & \dot{u}(r) = -s(\xi_c(r), \eta_c(r)) \\ (\mathrm{b}) & \dot{\xi}_c(r) = b_1(\xi_c(r), \eta_c(r)) \\ (\mathrm{c}) & \dot{\eta}_c(r) = b_2(\xi_c(r), \eta_c(r)) \end{cases} \tag{2.53}$$

where r is the parameter of the projected characteristic $(\xi_c(r), \eta_c(r))$, with initial condition

$$\begin{cases} (\mathrm{a}_0) & u(\xi_c(0), \eta_c(0)) = e^{g(\xi_0, \eta_0)} \\ (\mathrm{b}_0) & \xi_c(0) = \xi_0 \\ (\mathrm{c}_0) & \eta_c(0) = \eta_0 \end{cases}$$

with $(\xi_0, \eta_0) \in \partial\Omega$, has a unique solution.

3. Direct Surface Reconstruction Using Multiple Images. When using multiple light sources we have an additional requirement on the lighting directions. A further assumption needs to be made when more than two images are taken into account. By considering the

reflectance equation for Lambertian surfaces as a linear function with respect to the light source vector, we are constrained to consider only non-coplanar light sources. This inconvenience has been studied in [23] with respect to PS_3. It is well known that an image obtained with a linearly dependent light sources with respect to the other images does not add any additional information if shadows do not occur [31, 4]. In the following sections, we will focus our study on the orthographic case which can be easily extended to the perspective case by repeating similar passages. In the following part, we call the unknown as z even if considering orthographic viewing geometry. The extension to the perspective case follows trivially from analogue passages.

3.1. Weighted Photometric Stereo for multiple images with missing parts. If we have three linearly independent images obtained by individually shining three parallel-ray light sources in non-coplanar directions then we can consider the set of unique image pairs and have the following system of linear PDEs:

$$\begin{cases} b^{(1,2)}(x,y)\cdot\nabla z(x,y) = s^{(1,2)}(x,y) \\ b^{(1,3)}(x,y)\cdot\nabla z(x,y) = s^{(1,3)}(x,y) \\ b^{(2,3)}(x,y)\cdot\nabla z(x,y) = s^{(2,3)}(x,y) \end{cases} \tag{3.1}$$

of the same type as (2.10), where

$$\begin{aligned} b^{(h,k)}(x,y) = (I_k(x,y)\omega_1^h - I_h(x,y)\omega_1^k, \\ I_k(x,y)\omega_2^h - I_h(x,y)\omega_2^k) \end{aligned} \tag{3.2}$$

and

$$s^{(h,k)}(x,y) = I_k(x,y)\omega_3^h - I_h(x,y)\omega_3^k \tag{3.3}$$

with (h,k) being the combination of two of the first three natural integers without repetitions.

We can now describe our novel contribution which is to ensure the well posedness of the PS_n problem by exploiting the linearity of the basic differential formulation (2.10) and reducing it to a single PDE which can handle shadowed regions in a natural fashion. Since (2.10) does not lose the well-posedness if we multiply the equations by a function $q(x,y)$ on both sides (i.e. $b(x,y)$ and $f(x,y)$), we are able to define the ingredients of a weighted PS_n model (W-PS_n) by considering the functions:

$$b_n^w(x,y) = \sum_{p\in\binom{[n]}{2}} q_p(x,y)b^p(x,y) \tag{3.4}$$

and

$$s_n^w(x,y) = \sum_{p\in\binom{[n]}{2}} q_p(x,y)s^p(x,y) \tag{3.5}$$

where $\binom{[n]}{2}$ is the set of pairs of integer indices with no repetition. For example, if $n=3$ we have $\binom{[3]}{2} = \{(1,2),(1,3),(2,3)\}$.

The complete construction of the W-PS$_n$ formulation is therefore

$$\begin{cases} b_n^w(x,y)\cdot\nabla z(x,y) = s_n^w(x,y), & \text{a.e. } (x,y)\in\Omega \\ \hat{z}(x,y) = g(x,y) & \forall (x,y)\in\partial\Omega. \end{cases} \tag{3.6}$$

Next, we explain how shadows influence the definition of the weights (hence of b_n^w and s_n^w).

A key observation we can make is that it is possible to use weight-functions q_p that are vanishing while preserving the well-posedness of the problem. Obviously the main importance is related to the set of points where the functions q_p are null, not their signs.

Let us observe that the well-posedness of the differential formulation is guaranteed for image pixels lit in at least two images and preserved if the same condition holds in the multi-image, weighted case.

Since we want to exploit the Photometric Stereo technique, we assume that each pixel is illuminated in at least two images thereby avoiding reduction to a PS$_1$ problem. Our aim is to consider the weights as switches, able to locally nullify the involvement of an image pair in the summations (3.4) and (3.5) when the functions b^p and s^p for that pair do not contain relevant information due to the presence of shadows in the images involved. Since no ambient light is assumed in our set-up, we consider the point $(x,y)\in\overline{\Omega}$ shadowed in the ith image when $I_i(x,y)=0$. Now, by using the Heaviside function we can easily define the weights as follows:

$$q_{(h,k)} = H(I_h(x,y))H(I_k(x,y)). \tag{3.7}$$

These functions are not continuous and therefore in order to not complicate (3.6) by adding such discontinuous functions, we consider a regularization of (3.7). By assuming the shadows to be open sets, we can regularize the weights using *cutoff functions* as constructed in [14].

The uniqueness for the problem (3.6) can be shown following the same steps used for the two-image case. That is the following results hold:

1. (**Lemma 1**), the absence of a critical point for the projected characteristic field, i.e. $b_3^w(x,y)\neq(0,0)$;
2. (**Theorem 1**), the propagation of the information from the boundary is not prevented between two sets separated by discontinuity.

3.2. Some remarks on the uniqueness of solution. Let us emphasize the following symmetry property of the previous equations:

$$b^{(h,k)}(x,y)\cdot\nabla z(x,y) = f^{(h,k)}(x,y) \Leftrightarrow b^{(k,h)}(x,y)\cdot\nabla z(x,y) = f^{(k,h)}(x,y) \tag{3.8}$$

due to the structure of $b^{(h,k)}(x,y)$ and $f^{(h,k)}(x,y)$, obeying:

$$b^{(h,k)}(x,y) = -b^{(k,h)}(x,y) \quad \text{and} \quad f^{(h,k)}(x,y) = -f^{(k,h)}(x,y) \quad \forall (x,y)\in\Omega. \tag{3.9}$$

In this section we want to focus our attention on some issues regarding the sum of the three linear PDEs in (3.1). Let us sum them as follows:

$$\left(b^{(1,2)} - b^{(1,3)} + b^{(2,3)}\right)\cdot\nabla z(x,y) = \left(f^{(1,2)} - f^{(1,3)} + f^{(2,3)}\right) \tag{3.10}$$

which is the same as the equation below

$$\left(b^{(1,2)} + b^{(3,1)} + b^{(2,3)}\right)\cdot\nabla z(x,y) = \left(f^{(1,2)} + f^{(3,1)} + f^{(2,3)}\right) \tag{3.11}$$

simply using (3.8) and (3.9).

If we write explicitly the sum of the functions involved in the previous equations we obtain:

$$b^{(1,2)} + b^{(3,1)} + b^{(2,3)} = \widetilde{\omega}'(I_3 - I_2) + \widetilde{\omega}''(I_1 - I_3) + \widetilde{\omega}'''(I_2 - I_1) \tag{3.12}$$

and

$$f^{(1,2)} + f^{(3,1)} + f^{(2,3)} = \omega_3'(I_3 - I_2) + \omega_3''(I_1 - I_3) + \omega_3'''(I_2 - I_1). \tag{3.13}$$

Since every term is multiplied by the difference between one image and another, it is clear that both functions vanish simultaneously at all the points $(\widetilde{x}, \widetilde{y}) \in \overline{\Omega}$ for which:

$$I_1(\widetilde{x}, \widetilde{y}) = I_2(\widetilde{x}, \widetilde{y}) = I_3(\widetilde{x}, \widetilde{y}). \tag{3.14}$$

Obviously in this case the uniqueness result is compromised since Lemma 1 does not hold. We show now that the previous condition can be a concrete problem since the points $(\widetilde{x}, \widetilde{y})$ can be explicitly characterized in some illumination configurations. Let us consider a particular set up for the direction of the light sources. It is based on some optimality conditions studied in [33] where the light sources are taken in the same rings. In other words, comparing with our light sources definition, we have to take the unit vectors ω', ω'' and ω''' such that:

$$\omega_3' = \omega_3'' = \omega_3'''. \tag{3.15}$$

Now the special points $(\widetilde{x}, \widetilde{y})$ can easily be found depending on the surface z. In fact, the following system:

$$\begin{aligned} I_3(\widetilde{x}, \widetilde{y}) - I_2(\widetilde{x}, \widetilde{y}) = \frac{-\nabla z(\widetilde{x}, \widetilde{y}) \cdot \widetilde{\omega}''' + \omega_3'''}{\sqrt{1 + ||\nabla z||^2}} - \frac{-\nabla z(\widetilde{x}, \widetilde{y}) \cdot \widetilde{\omega}'' + \omega_3''}{\sqrt{1 + ||\nabla z||^2}} = \\ \frac{-\nabla z(\widetilde{x}, \widetilde{y}) \cdot \widetilde{\omega}''' + \nabla z(\widetilde{x}, \widetilde{y}) \cdot \widetilde{\omega}''}{\sqrt{1 + ||\nabla z||^2}} = 0 \end{aligned} \tag{3.16}$$

$$\begin{aligned} I_1(\widetilde{x}, \widetilde{y}) - I_3(\widetilde{x}, \widetilde{y}) = \frac{-\nabla z(\widetilde{x}, \widetilde{y}) \cdot \widetilde{\omega}' + \omega_3'}{\sqrt{1 + ||\nabla z||^2}} - \frac{-\nabla z(\widetilde{x}, \widetilde{y}) \cdot \widetilde{\omega}''' + \omega_3'''}{\sqrt{1 + ||\nabla z||^2}} = \\ \frac{-\nabla z(\widetilde{x}, \widetilde{y}) \cdot \widetilde{\omega}' + \nabla z(\widetilde{x}, \widetilde{y}) \cdot \widetilde{\omega}'''}{\sqrt{1 + ||\nabla z||^2}} = 0 \end{aligned} \tag{3.17}$$

$$\begin{aligned} I_2(\widetilde{x}, \widetilde{y}) - I_1(\widetilde{x}, \widetilde{y}) = \frac{-\nabla z(\widetilde{x}, \widetilde{y}) \cdot \widetilde{\omega}'' + \omega_3''}{\sqrt{1 + ||\nabla z||^2}} - \frac{-\nabla z(\widetilde{x}, \widetilde{y}) \cdot \widetilde{\omega}' + \omega_3'}{\sqrt{1 + ||\nabla z||^2}} = \\ \frac{-\nabla z(\widetilde{x}, \widetilde{y}) \cdot \widetilde{\omega}'' + \nabla z(\widetilde{x}, \widetilde{y}) \cdot \widetilde{\omega}'}{\sqrt{1 + ||\nabla z||^2}} = 0 \end{aligned} \tag{3.18}$$

holds if and only if $\nabla z(\widetilde{x}, \widetilde{y}) = 0$ since $\widetilde{\omega}' \neq \widetilde{\omega}'' \neq \widetilde{\omega}'''$ is the main assumption for the PS technique. This means that the differential equation (3.10) (together with the same boundary condition of (3.1)) does not have a unique smooth solution if it admits stationary points.

It is clear that in fact, the set of the points where $\left(b^{(1,2)} - b^{(1,3)} + b^{(2,3)}\right)$ (or $\left(f^{(1,2)} - f^{(1,3)} + f^{(2,3)}\right)$) vanishes can also be used to find the stationary points of the unknown surface z. In this section, we obtained results comparable with [8], and merging both theories would be possible for future works.

3.3. Numerical stability in presence of images with noise. As seen in the previous section, the way to solve the differential problem requires to propagate the information starting from the boundary condition. Exploiting the characteristic strip expansion it is possible to give a constructive proof of the existence of a unique Lipschitz solution. Since we consider numerical schemes that mimicing the characteristic method, it is important to study the convergence analysis for a general one step numerical scheme solving the characteristic ODE system perturbed for a noisy image. Let us then consider the following Cauchy problem (i.e. characteristic problem):

$$\begin{cases} \text{(a)} & \dot{x}_c(s) = b_3^{w,1}(x_c(s), y_c(s)) \\ \text{(b)} & \dot{y}_c(s) = b_3^{w,2}(x_c(s), y_c(s)) \\ \text{(c)} & \dot{\hat{z}}(s) = s_3^{w}(x_c(s), y_c(s)) \end{cases} \tag{3.19}$$

where s is the variable of the parameterization and $\hat{z}(s)$ is the value of the surface on the projected characteristic (that is $\hat{z}(s) = \hat{z}(x_c(s), y_c(s))$).

The initial condition to integrate the previous system of ordinary differential equation is taken from the values of the function z known on the boundary

$$\Gamma_{in} = \{(\widetilde{x}, \widetilde{y}) \in \partial\Omega : \nu(\widetilde{x}, \widetilde{y}) \cdot \lim_{\substack{(x,y)\to(\widetilde{x},\widetilde{y}) \\ (x,y)\in\Omega}} b_3^w(x, y) \leq 0\} \tag{3.20}$$

where $\nu(x, y)$ represents the outgoing normal to the boundary $\partial\Omega$. It is clear that, in the previous definition, the limit is taken, because a point of discontinuity may belong to the boundary. Then

$$\begin{cases} \text{(a}_0\text{)} & x_c(0) = x_0 \\ \text{(b}_0\text{)} & y_c(0) = y_0 \\ \text{(c}_0\text{)} & \hat{z}(x_c(0), y_c(0)) = g(x_0, y_0) \end{cases}$$

with $(x_0, y_0) \in \partial\Omega$.

Let us consider b_3^w, s_3^w perturbed as follows:

$$\begin{aligned} \overline{b}_3^{w,1} &= q_{(1,2)}\big((I_2+\eta_2)\omega_1' - (I_1+\eta_1)\omega_1''\big) + \\ &\quad q_{(1,3)}\big((I_3+\eta_3)\omega_1' - (I_1+\eta_1)\omega_1'''\big) + q_{(2,3)}\big((I_3+\eta_3)\omega_1'' - (I_2+\eta_2)\omega_1'''\big) \\ \overline{b}_3^{w,2} &= q_{(1,2)}\big((I_2+\eta_2)\omega_2' - (I_1+\eta_1)\omega_2''\big) + \\ &\quad q_{(1,3)}\big((I_3+\eta_3)\omega_2' - (I_1+\eta_1)\omega_2'''\big) + q_{(2,3)}\big((I_3+\eta_3)\omega_2'' - (I_2+\eta_2)\omega_2'''\big) \\ \overline{s}_3^{w} &= q_{(1,2)}\big((I_2+\eta_2)\omega_3' - (I_1+\eta_1)\omega_3''\big) + \\ &\quad q_{(1,3)}\big((I_3+\eta_3)\omega_3' - (I_1+\eta_1)\omega_3'''\big) + q_{(2,3)}\big((I_3+\eta_3)\omega_3'' - (I_2+\eta_2)\omega_3'''\big) \end{aligned} \tag{3.21}$$

where η_1, η_2 and η_3 are the additive noise at point $(x_c(s), y_c(s))$.

For simplicity, let us study the three lighted point situation, that is where $q_{(1,2)} = q_{(1,3)} = q_{(2,3)} = 1$. Exploiting the linearity of the images (i.e. of the noise) in the previous quantities, we have:

$$\begin{aligned} \overline{b}_3^{w,1} &= b_3^{w,1} - \eta_1(\omega_1'' + \omega_1''') + \eta_2(\omega_1' - \omega_1''') + \eta_3(\omega_1' + \omega_1'') \\ \overline{b}_3^{w,2} &= b_3^{w,2} - \eta_1(\omega_2'' + \omega_2''') + \eta_2(\omega_2' - \omega_2''') + \eta_3(\omega_2' + \omega_2'') \\ \overline{s}_3^{w} &= s_3^{w} - \eta_1(\omega_3'' + \omega_3''') + \eta_2(\omega_3' - \omega_3''') + \eta_3(\omega_3' + \omega_3'') \end{aligned} \tag{3.22}$$

that is the additive noise of each image brings us to consider an additive perturbation of (2.53). Rewriting (3.19) in a compact form, we have

$$\begin{cases} \dot{Z}(s) = k(s) + \eta \\ \dot{Z}(0) = g(x_0, y_0) + \eta_0 \end{cases} \tag{3.23}$$

where Z and k take values in $\mathbb{R}^3$ and η_0 is the error on the boundary condition. Let us emphasize that, since (3.19) admits a unique solution, we can assume k to be a Lipschitz function. Let us recall that the strategy in order to prove the uniqueness of a weak solution is to consider the characteristic method from the boundary till the first curve of discontinuity $\gamma(t)$ and to start again with another boundary problem. Of course, the Lipschitz property of k has to be taken into account locally, excluding the points of discontinuity for k.

Let us focus our attention on a general way to write a one step numerical scheme which solves (3.23):

$$\begin{cases} u_{n+1} = u_n + \Delta\Phi(s_n, u_n, k(s_n), \Delta) + \eta_n \\ u_0 = g(x_0, y_0) + \eta_0 \end{cases} \tag{3.24}$$

where Δ is the uniform discretization step and Φ is a general function of discretization for the first derivative. As usual, $u_n = u(s_n)$ is the approximation of the solution $Z_n = Z(s_n)$ and $\eta_n = \eta(s_n)$ is the perturbation at point s_n. We want to provide an estimation of the error at point s_{n+1}, that is:

$$e_{n+1} = Z_{n+1} - u_{n+1} = (Z_{n+1} - u^*_{n+1}) + (u^*_{n+1} - u_{n+1}) \tag{3.25}$$

where u^*_{n+1} is the error computing the value of the approximation starting from correct data (see Figure 3.1).

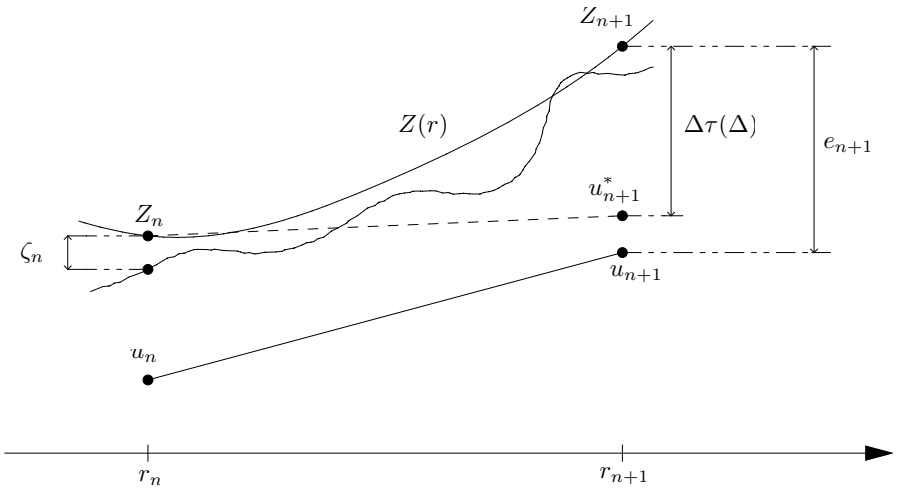


Figure 3.1. *Geometric interpretation of the local and global error in s_{n+1} in a perturbed configuration.*

We have:

$$Z_{n+1} - u^*_{n+1} = \Delta\tau(\Delta) \tag{3.26}$$

which represents the consistency error of the method plus the perturbation in s_{n+1}.

The other part of the right hand side of (3.25) becomes

$$\begin{aligned} u^*_{n+1} - u_{n+1} =& Z_n + \Delta\Phi(s_n, Z_n, k(s_n), \Delta) - u_n - \Delta\Phi(s_n, u_n, k(s_n), \Delta) - \eta_n = \\ & e_n + \Delta\big[\Phi(s_n, Z_n, k(s_n), \Delta) - \Phi(s_n, u_n, k(s_n), \Delta)\big] - \eta_n \end{aligned} \tag{3.27}$$

that is we have

$$\begin{aligned} ||e_{n+1}|| \leq & \Delta||\tau(\Delta)|| + ||\eta_n|| + ||e_n|| + \Delta||\Phi(s_n, Z_n, k(s_n), \Delta) - \Phi(s_n, u_n, k(s_n), \Delta)|| \\ & \Delta||\tau(\Delta)|| + ||\eta_n|| + ||e_n|| + \Delta L||Z_n - u_n|| = \Delta||\tau(\Delta)|| + ||\eta_n|| + (1 + \Delta L)||e_n|| \end{aligned} \tag{3.28}$$

exploiting the local Lipschitzianity of Φ (i.e. dependence on the numerical schemes) induced by the k local regularity. For example if Φ represents the Euler discretization then L is exactly the Lipschitz constant of k.

Repeating the same reasoning for $||e_n||$ and so on, we have the following chain

$$\begin{gathered} ||e_{n+1}|| \leq \Delta||\tau(\Delta)||\big[1 + (1 + \Delta L) + (1 + \Delta L)^2 + \ldots + (1 + \Delta L)^n\big] + \\ ||\eta_n|| + (1 + \Delta L)||\eta_{n-1}|| + (1 + \Delta L)^2||\eta_{n-2}|| + \ldots + (1 + \Delta L)^{n-1}||\eta_1|| + (1 + \Delta L)^n||\eta_0|| \leq \\ \big[\Delta||\tau(\Delta)|| + \Theta\big](1 + (1 + \Delta L) + (1 + \Delta L)^2 + \ldots + (1 + \Delta L)^n) = \\ \tfrac{(1+\Delta L)^{n+1}-1}{hL}\big[\Delta||\tau(\Delta)|| + \Theta\big] \end{gathered} \tag{3.29}$$

where $\Theta = \max_{0 \leq i \leq n} ||\eta_i||$. Now, since $1 + \Delta L \leq e^{\Delta L}$ and $(n + 1)\Delta = s_{n+1} - s_0$, we can continue the estimation as follows:

$$\begin{aligned} ||e_{n+1}|| \leq \frac{e^{L(s_{n+1}-s_0)} - 1}{hL}\big[\Delta||\tau(\Delta)|| + \Theta\big] \leq \\ e^{L(s_{n+1}-s_0)}\frac{1}{L}\left(||\tau(\Delta)|| + \frac{\Theta}{\Delta}\right) - \frac{1}{\Delta L}\big[\Delta||\tau(\Delta)|| + \Theta\big] \leq \\ e^{L(s_{n+1}-s_0)}\frac{1}{L}\left(||\tau(\Delta)|| + \frac{\Theta}{\Delta}\right). \end{aligned} \tag{3.30}$$

It follows that we cannot have the convergence when $\Delta \to 0$ because of the presence of the noise. Of course, there exists an optimal step for which the global error is minimum. Under this value the noise effects become dominant in the truncation error.

4. Numerical Schemes.

Next, we consider the numerical methods approximating the solution on the image domain restricted to the set $[-1, 1]^2$ with a discretization space step Δ_x and Δ_y, [22]. We use them because they converge very fast and are easy to implement.

4.1. Forward numerical schemes.

We want to recall now the numerical schemes used for the forward approximation of (2.7) where the propagation of the information starts from the inflow part of the boundary Γ_{in}. We will use the orthographic variables (x, y), but the formulation can be easily reinterpreted for the perspective case just using (ξ, η). Furthermore, simplify the notation by denoting $b(x_i, y_j)$ by $b_{i,j} = (b^1_{i,j}, b^2_{i,j})$ and $s(x_i, y_j)$ by $s_{i,j}$.

Forward upwind Scheme:

$$R^F_{i,j} = \frac{\Delta_y |b^1_{i,j}| R^F_{i-\mathrm{sgn}(b^1_{i,j}),j} + \Delta_x |b^2_{i,j}| R^F_{i,j-\mathrm{sgn}(b^2_{i,j})} + \Delta_x \Delta_y s_{i,j}}{|b^1_{i,j}|\Delta_y + |b^2_{i,j}|\Delta_x}. \tag{4.1}$$

In our case the numerical schemes are applied to digital images where clearly $\Delta_x = \Delta_y = \Delta$.

Forward semi-Lagrangian scheme:

$$r^F_{i,j} = r^F(x_i - h\alpha^1_{i,j}, y_j - h\alpha^2_{i,j}) + \frac{s_{i,j}}{|b_{i,j}|}h \tag{4.2}$$

where $\alpha_{i,j} = \frac{b_{i,j}}{|b_{i,j}|}$ and the parameter $h > 0$ is assumed equal to the size of the grid Δ in order to reach the highest order of convergence equal to one ([22]).

4.2. Backward numerical schemes. The backward numerical schemes are based on approximation of the surface by propagating the information stored on the outflow part of the boundary

$$\Gamma_{out} = \partial\Omega \setminus \Gamma_{in}. \tag{4.3}$$

Backward upwind Scheme:

$$R^B_{i,j} = \frac{\Delta_y |b^1_{i,j}| R^B_{i+\mathrm{sgn}(b^1_{i,j}),j} + \Delta_x |b^2_{i,j}| R^B_{i,j+\mathrm{sgn}(b^2_{i,j})} - \Delta_x \Delta_y s_{i,j}}{|b^1_{i,j}|\Delta_y + |b^2_{i,j}|\Delta_x}. \tag{4.4}$$

Backward semi-Lagrangian scheme:

$$r^B_{i,j} = r^B(x_i + h\alpha^1_{i,j}, y_j + h\alpha^2_{i,j}) - \frac{s_{i,j}}{|b_{i,j}|}h. \tag{4.5}$$

5. W-PS$_n$ With No Boundary Condition. In Sections 2 and 3 we extended the PS models by assuming knowledge of the boundary condition $g(x, y)$. Clearly such a hypothesis compromises the use of that model in most real applications. It is therefore important to find a way to solve the PS$_n$ problem without requiring knowledge of the boundary condition.

To solve this problem we design a numerical strategy which involves selecting a single arbitrarily valued initial seed point within the reconstruction domain and robustly manipulating the path of the characteristics. We do this in order to numerically integrate the linear differential problem (3.6) so as to let the information travel in the most convenient directions for the whole domain.

5.1. Controlling the characteristic field. On the way to defining a numerical strategy we will need to manipulate the path along which the information travels. To do this we will employ the following result:

Theorem *Let $b^p(x, y)$ be the vector field of (3.2) where $p \in \binom{[n]}{2}$. Then, $\forall p_1, p_2 \in \binom{[n]}{2}$ and $\forall (x, y) \in \Omega$ we have:*

$$b^{p_1}(x, y) \cdot b^{p_2}(x, y) \neq \pm |b^{p_1}(x, y)||b^{p_2}(x, y)|. \tag{5.1}$$

Proof

For simplicity, let us fix the indices p_1 and p_2 as $(1,2)$ and $(1,3)$ respectively. In order to prove by contradiction that $b^{(1,2)}$ and $b^{(1,3)}$ are never parallel, we assume that there exists a point $(\widetilde{x},\widetilde{y}) \in \Omega$ such that:

$$b^{(1,2)}(\widetilde{x},\widetilde{y}) \cdot b^{(1,3)}(\widetilde{x},\widetilde{y}) = \pm |b^{(1,2)}(\widetilde{x},\widetilde{y})||b^{(1,3)}(\widetilde{x},\widetilde{y})|. \tag{5.2}$$

For the sake of clarity we omit the dependence on $(\widetilde{x},\widetilde{y})$. Now, by squaring both sides we have:

$$\left[b_1^{(1,2)}b_1^{(1,3)} + b_2^{(1,2)}b_2^{(1,3)}\right]^2 = \left[\left(b_1^{(1,2)}\right)^2 + \left(b_2^{(1,2)}\right)^2\right]\left[\left(b_1^{(1,3)}\right)^2 + \left(b_2^{(1,3)}\right)^2\right] \tag{5.3}$$

and by writing $b^{(1,2)}$ and $b^{(1,3)}$ explicitly from (3.2) we get:

$$\begin{aligned}\left[I_2I_3(\omega_1')^2 - I_1I_2\omega_1'\omega_1''' - I_1I_3\omega_1'\omega_1'' + I_1^2\omega_1''\omega_1''' + I_2I_3(\omega_2')^2 - I_1I_2\omega_2'\omega_2''' - I_1I_3\omega_2'\omega_2'' + I_1^2\omega_2''\omega_2'''\right]^2 = \\ \left[(I_2\omega_1' - I_1\omega_1'')^2 + (I_2\omega_2' - I_1\omega_2'')^2\right]\left[(I_3\omega_1' - I_1\omega_1''')^2 + (I_3\omega_2' - I_1\omega_2''')^2\right]\end{aligned} \tag{5.4}$$

Now, let us write the reflectance function by simplifying the notation as follows:

$$I_j = \rho(x,y)\frac{i_j(x,y)}{\sqrt{1+|\nabla z(x,y)|^2}}, \quad j = 1,2,3 \tag{5.5}$$

and substitute them into (5.4). We note that the quantity

$$\frac{\rho(x,y)}{\sqrt{1+|\nabla z(x,y)|^2}} \tag{5.6}$$

is nonvanishing and therefore can be eliminated from both sides of (5.4). Finally we can write (5.4) as follows:

$$\begin{aligned}\left[i_2i_3(\omega_1')^2 - i_1i_2\omega_1'\omega_1''' - i_1i_3\omega_1'\omega_1'' + i_1^2\omega_1''\omega_1''' + i_2i_3(\omega_2')^2 - i_1i_2\omega_2'\omega_2''' - i_1i_3\omega_2'\omega_2'' + i_1^2\omega_2''\omega_2'''\right]^2 = \\ \left[(i_2\omega_1' - i_1\omega_1'')^2 + (i_2\omega_2' - i_1\omega_2'')^2\right]\left[(i_3\omega_1' - i_1\omega_1''')^2 + (i_3\omega_2' - i_1\omega_2''')^2\right]\end{aligned} \tag{5.7}$$

and after some algebraic manipulation we get:

$$i_1(\omega_1'''\omega_2''\omega_3' - \omega_1''\omega_2'''\omega_3' - \omega_1'''\omega_2'\omega_3'' + \omega_1'\omega_2'''\omega_3'' + \omega_1''\omega_2'\omega_3''' - \omega_1'\omega_2''\omega_3''') = 0. \tag{5.8}$$

Assuming that we are considering a non shadowed point for the first image (i.e. $i_1 > 0$), we have that (5.8) is satisfied only if the light sources are collinear since it is equivalent to the following:

$$\det\begin{pmatrix}\omega_1' & \omega_2' & \omega_3' \\ \omega_1'' & \omega_2'' & \omega_3'' \\ \omega_1''' & \omega_2''' & \omega_3'''\end{pmatrix} = 0 \tag{5.9}$$

which is in contradiction with the Photometric Stereo assumption.

□

In other words this theorem says that two different vector fields b^{p_1} and b^{p_2} cannot be parallel. We will use this fact to control the direction of the characteristics for the case when

$n = 3$ which can be easily generalised to any number of images. Let us introduce the set of three-lighted pixels $\mathcal{M}$ through its indicator function as follows:

$$\mathbb{1}_{\mathcal{M}}(x, y) = H(I_1(x, y))H(I_2(x, y))H(I_3(x, y)). \tag{5.10}$$

We would like to control the summation functions q_p in (3.4) and (3.5) such that we can take a linear weighted combination as given by the following two indexes:

$$(p_1^*, p_2^*) = argmin_{(p_1,p_2)\in\binom{[3]}{2}} |b^{p_1} \cdot b^{p_2}| \tag{5.11}$$

with the aim to span the set of all possible directions of the derivatives by using the two least ill-conditioned directions $b^{p_1^*}$ and $b^{p_2^*}$. This is permitted due to the linearity of the basic differential formulation (2.10).

Since $q_p(x, y) \equiv 1$, $\forall (x, y) \in \mathcal{M}$ and $\forall p \in \binom{[3]}{2}$, we can sum the equations in (3.1) as follows:

$$\left(\alpha b^{p_1^*} + \beta b^{p_2^*}\right) \cdot \nabla z = \alpha f^{p_1^*} + \beta f^{p_2^*} \tag{5.12}$$

where α and β are real coefficients.

We can now fully control the direction in which we compute the first derivatives of z independently at any pixel provided no shadows are involved. We will choose (α, β) in order to control the characteristic direction at a pixel to be in the most favourable direction as required by some integration strategy. On an axis aligned discretization grid we define eight primary directions of integration (Figure 5.1).

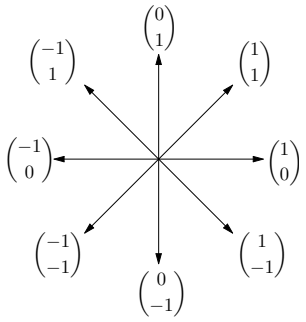


Figure 5.1. *The main directions for the integration strategy on an axis aligned grid.*

We recognize these as the integration directions resulting from the possible locations accessible by the numerical schemes derived in the next section. Once we have chosen a particular direction d at a point $(x, y) \in \mathcal{M}$ we can compute values for (α, β) so that

$$\begin{pmatrix} b_1^{p_1^*}(x, y) & b_1^{p_2^*}(x, y) \\ b_2^{p_1^*}(x, y) & b_2^{p_2^*}(x, y) \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} d_1(x, y) \\ d_2(x, y) \end{pmatrix}. \tag{5.13}$$

Provided that the light sources used to produce the three images are non-coplanar then Theorem (5.1) guarantees that the vectors $b^{p_1^*}$ and $b^{p_2^*}$ are not parallel and thus this 2×2 system is guaranteed to have a unique solution. Furthermore by virtue of (5.11) we have ensured that

it is well conditioned. Thus for each point $(x, y) \in \mathcal{M}$ we can compute coefficients (α, β) such that

$$\left(\alpha b^{p_1^*} + \beta b^{p_2^*}\right) \cdot \nabla z = \alpha f^{p_1^*} + \beta f^{p_2^*}, \tag{5.14}$$

which is now a pixel specific version of (3.6).

5.2. Integration strategy for W-PS$_3$. We aim to minimize the accumulation of error during the numerical integration and therefore define a strategy for recovering the surface without the use of a-priori boundary conditions.

We have the following steps inspired by the wavefront expansion principle of fast marching methods over flat domains:

1. fix an arbitrary value to z for a point towards the center of the image domain not in shadow, namely $(\bar{x}, \bar{y})$ (see the orange point in Figure 5.2) and add all of that point's neighbors to a list of pixels to be visited;
2. traverse the list of pixels to be visited and update the value for z for each one by calculating p_1^*, p_2^*, α and β after determining what information is available as required by the forward schemes (4.1) or (4.2) derived in the next section;
3. for each newly visited pixel add its unvisited neighbors to the list of pixels to be visited;
4. in case of shadows we can change the wavefront propagation direction in order to *surround* the shadow sets (i.e. computing the boundary condition) as shown by the red arrows in Figure 5.2 and then solve the appropriate equation in (3.1) in the set of shadowed pixels when the wavefront expansion direction is in agreement with the shadowed pixel's characteristic direction;
5. the above steps are repeated until some stopping condition on convergence is fulfilled.

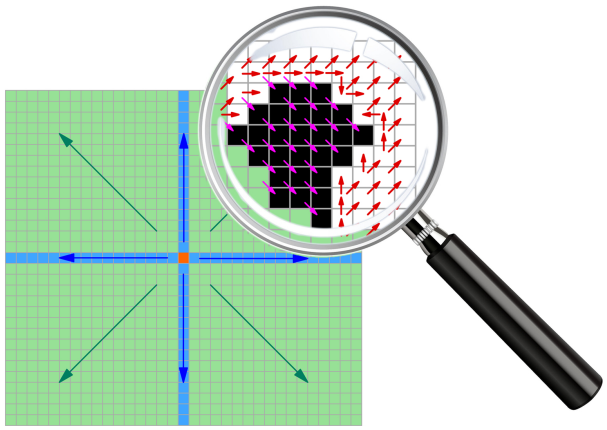


Figure 5.2. *We fix an arbitrary point (in orange) and perform surface recovery along an expanding wavefront of visited pixels. When shadows occur, we orient the vector field in order to avoid them when their directions are inconsistent with the direction of expansion of the wavefront.*

The main idea that we are proposing is that for all nonshadow pixels we can orient the characteristic field so that its direction is convenient for use in our integration strategy. This is only possible using our new formulation in (3.6). In this way the advancing wavefront constantly makes new pixels available to be updated even when they are in the shadow set

because at some point the direction of advance of the wavefront will be consistent with the direction of a pixel's characteristic which sits along the border of the shadow set. This pixel is then added to the set of pixels to be visited and in turn will act as a seed pixel that enables other points in the shadow set to be updated in the next incremental advance of the wavefront. In such a way the entire set of pixels can be visited and correctly updated.

6. Numerical Tests. This section describes the experiments conducted with the proposed numerical schemes: the semi-Lagrangian and the upwind finite difference scheme, each in its forward and backward formulations.

For the numerical tests we utilized two surfaces, each with a different geometrical and analytical characteristics. The surface (Fig. 6.1 left):

$$\hat{z}_{reg}(x,y) = \begin{aligned} &10 + 0.1\sin(x^2+y^2) - e^{-((3(x-0.2))^2+(3(y+0.2))^2)} \\ &+e^{-((3(x+1.6))^2+(3(y-2.1))^2)} + e^{-((3(x+0.9))^2+(3(y+1.1))^2)} \end{aligned}$$

($\hat{z}(x,y) \in [9.0110, 11.088]$) is a regular surface with three extremes that guarantee some very steep slopes. The surface (Fig. 6.1 center):

$$\hat{z}_{irreg}(x,y) = \begin{aligned} &10 - |0.4\sin(x^2+y^2) - e^{-((3(x-0.2))^2+(3(y+0.2))^2)} \\ &+e^{-((3(x+1.3))^2+(3(y-1.4))^2)} + e^{-((3(x+0.9))^2+(3(y+1.1))^2)}| \end{aligned}$$

($\hat{z}(x,y) \in [8.635, 10]$) has some curves of non differentiability. In particular, its boundary is nondifferentiable everywhere. The surface (Fig. 6.1 right):

For each of these surfaces, we computed the perspective image under two light source directions according to the procedure described by Tankus et al. [39] (Fig. 6.2). We used a constant focal length $f = 1$ for all images. To demonstrate reconstruction under a nonconstant, discontinuous albedo function we artificially generated dark stripes in the images (see Fig. 6.2). These stripes are of albedo $\rho = 0.5$, whereas $\rho = 1$ in all other parts of the test images. We repeated the experiements with images of several sizes: 100×100, 200×200, 400×400, and 800×800 pixels.

Reconstruction by the suggested perspective semi-Lagrangian and upwind schemes is highly accurate, as demonstrated in Fig. 6.3. We see that the suggested perspective methods faithfully reconstructed the surfaces, despite the irregularities.

Let us first show that the numerical schemes converge despite the irregularities, and examine their rates of convergence. Tables 6.1 and 6.2 examine the convergence rate of the numerical schemes using the L^∞ norm in the perspective coordinate system $(\xi, \eta, z(\xi,\eta))$. Indeed, both perspective schemes converge linearly (i.e., with order 1), because doubling the number of grid nodes halves the error. Thus, the theoretical convergence can be obtained in practice, and in a relatively short running time.

Having demonstrated convergence, let us now explore the accuracy of surface reconstruction. This is quantified by the root mean square error (RMSE). We compute RMSE in both the perspective coordinate system $(\xi, \eta, z(\xi,\eta))$, and the real-world coordinate system $(x, y, \hat{z}(x,y))$. The RMSE measures have the same units as surface depth, and can therefore be directly compared to the aforementioned ranges of $\hat{z}(x,y)$ values. As the original depth

values are around 10 and reconstruction error is generally less than 0.1 (and sometimes even two orders of magnitude less than this), the mean error is less than 1% in these examples, generally speaking.

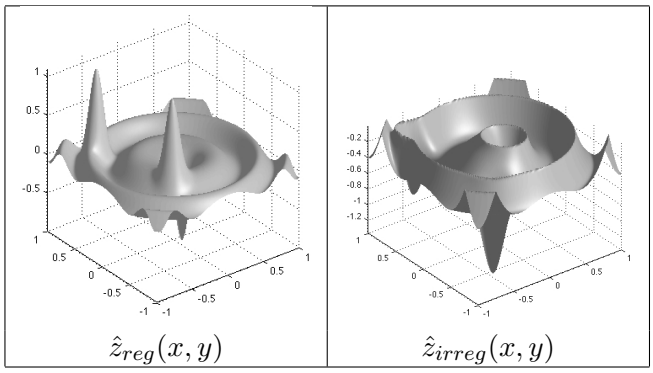


Figure 6.1. *The surfaces used for the numerical tests, each with different geometrical and analytical properties.*

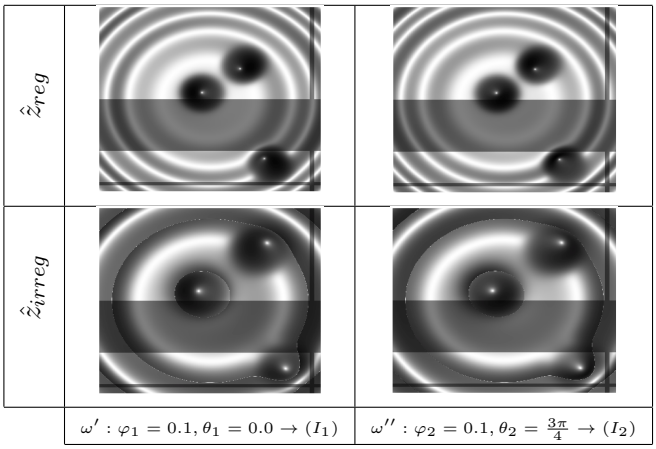


Figure 6.2. *Perspective images of the respective surfaces of Fig. 6.1, used as inputs to the algorithms. The light source directions appear on the bottom:* $\omega = (\sin(\varphi)\cos(\theta), \sin(\varphi)\sin(\theta), \cos(\varphi))$. *The albedo on the dark stripe of each image is* $\rho = 0.5$*; otherwise,* $\rho = 1$.

6.1. Numerical Tests for the Direct Reconstruction. In the first part of the numerical tests we show a numerical analysis of convergence of the numerical schemes introduced in the previous section. In this synthetic case we take into account a Lipschitz surface that, besides points of discontinuity, presents high slope (i.e. a large Lipschitz constant). The initial images shown in Figure 6.4 are synthesized using an analytical function of a surface z in the domain $\overline{\Omega}_d = [-1, 1]^2$.

The images include missing image regions together with a non constant albedo mask. We corrupt the image data with 5% Gaussian noise to further improve the realism of the simulation. Table 6.3 displays the error together with the measured run time at convergence. For this work the parallelism available to an implementation of the integration strategy has not

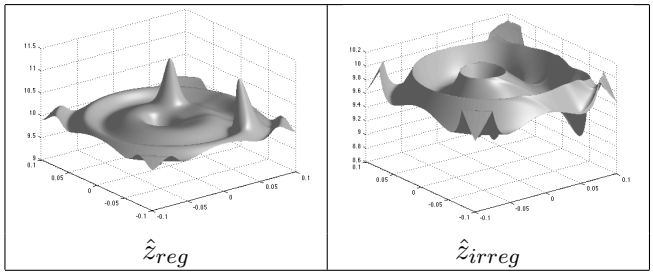


Figure 6.3. *Reconstruction by the proposed perspective backward semi-Lagrangian scheme, using the input images of Fig. 6.2. The surfaces are flipped compared to the original ones (Fig. 6.1) because of the perspective projection. The original surfaces were faithfully recovered.*

been utilized. Despite this limitation the algorithm runs at 6 *fps* for images of size 500×500. It should be noted though that this is a synthetic case and clearly the presence of noise influences the convergence of the numerical schemes and so they do not preserve the same rate of convergence with respect to the images without noise and thus have longer runtimes. Figure 6.5 shows the surfaces recovered from noisy images with four megapixels. Even if the reconstruction accurately preserves the shape of the surface, it produces some artifacts due to the noise and the rectangular occlusions. In this algorithm no regularisation procedure has been adopted in order to make the surface smoother.

		semi-lag Forward			upwind Forward		
	Δ	L^∞	MSE-persp	MSE-real	L^∞	MSE-persp	MSE-real
$\hat{z}_{reg}$	100	7.582×10^{-1}	0.079965	0.08006	6.780×10^{-1}	0.073121	0.073157
	200	3.543×10^{-1}	0.048369	0.04849	3.245×10^{-1}	0.046625	0.046775
	400	1.733×10^{-1}	0.027498	0.027577	1.631×10^{-1}	0.02839	0.028495
	800	8.567×10^{-2}	0.014932	0.014977	8.121×10^{-2}	0.016325	0.016389
$\hat{z}_{irreg}$	100	6.726×10^{-1}	0.11174	0.10957	4.693×10^{-1}	0.11507	0.11299
	200	4.977×10^{-1}	0.067081	0.068095	3.925×10^{-1}	0.078578	0.080503
	400	3.381×10^{-1}	0.037985	0.039874	2.664×10^{-1}	0.051528	0.054701
	800	2.174×10^{-1}	0.020888	0.02254	1.590×10^{-1}	0.035682	0.038287

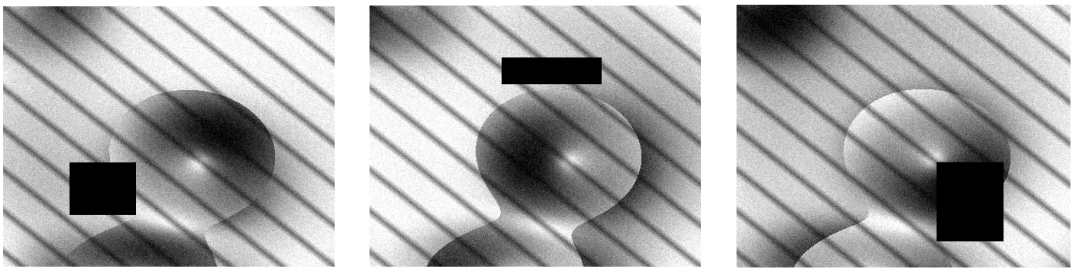
Table 6.1

Convergence and accuracy of the forward numerical schemes for each surface of Fig. 6.1. For each surface we examined images of size $\Delta \times \Delta$ pixels, and computed three error measures: the L^∞ norm in the perspective coordinate system, RMSE in the perspective system, and RMSE in the real-world coordinate system. The L^∞ norm shows that convergence is linear (i.e., order 1). The RMSE measures quantify the accurate reconstruction with respect to the original surface.

		semi-lag Backward			upwind Backward		
	Δ	L^∞	MSE-persp	MSE-real	L^∞	MSE-persp	MSE-real
$\hat{z}_{reg}$	100	7.582×10^{-1}	0.024478	0.025474	2.399×10^{-1}	0.0094924	0.011658
	200	1.789×10^{-1}	0.013978	0.014514	9.918×10^{-2}	0.0061304	0.007485
	400	8.494×10^{-2}	0.007591	0.0078847	4.662×10^{-2}	0.003698	0.0045026
	800	4.113×10^{-2}	0.003997	0.0041618	2.279×10^{-2}	0.0021912	0.0026803
$\hat{z}_{irreg}$	100	2.929×10^{-1}	0.040556	0.042861	1.673×10^{-1}	0.029009	0.028976
	200	2.952×10^{-1}	0.020974	0.023794	1.025×10^{-1}	0.019232	0.019162
	400	2.014×10^{-1}	0.012261	0.014336	8.878×10^{-2}	0.015983	0.016061
	800	1.697×10^{-1}	0.0072217	0.0086446	8.930×10^{-2}	0.014743	0.015029

Table 6.2

Convergence and accuracy of the backward numerical schemes for each surface of Fig. 6.1. The table is organized similarly to Table 6.1. The rate of convergence of the backward algorithm is the same as of the forward schemes (order 1). Accurate reconstruction is also achieved by the backward schemes.



I_1 I_2 I_3

Figure 6.4. *Set of synthetic images used in which the albedo is composed of diagonal stripes. 5% Gaussian noise and exaggerated black patches representing shadows are used to corrupt the images.*

		up-wind		semi-Lagrangian	
	Δ	L^∞	time (sec)	L^∞	time (sec)
5 %	500	3.539×10^{-2}	0.162	2.332×10^{-2}	0.171
	1000	2.185×10^{-2}	0.561	1.166×10^{-2}	0.623
	2000	1.368×10^{-2}	2.783	6.248×10^{-3}	3.029
	500	6.635×10^{-2}	0.161	5.855×10^{-2}	0.274
	1000	3.578×10^{-2}	0.553	3.698×10^{-2}	1.201
	2000	3.917×10^{-2}	2.684	3.916×10^{-2}	4.354

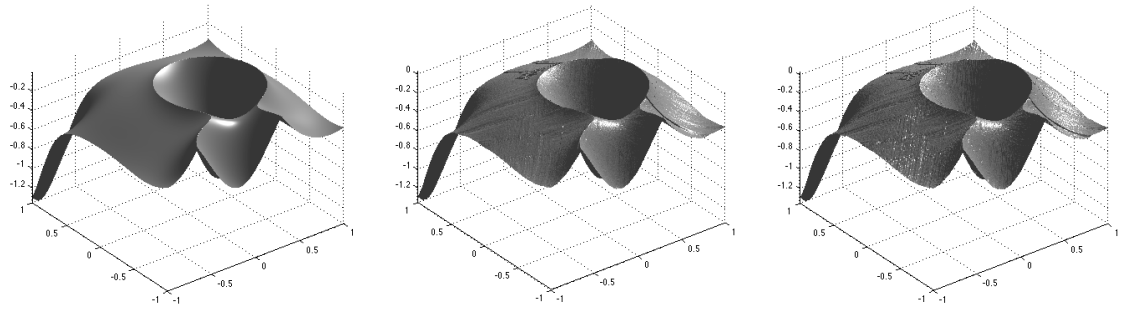
Table 6.3

In order to show that the order of convergence of the numerical schemes is one, we compute the L^∞ error doubling the size of the images (adding also 5% of Gaussian noise) starting from data of 500×500 to 2000×2000 pixels.

In the first row of Figure 6.6 we consider the problem of the reconstruction of the well-known Beethoven bust after complicating the shape recovery by adding black occlusions. As shown in Figure 6.7 the computed shape does not have evident artifacts also in zones where the information has been removed.

As a last example we consider the reconstruction of a coin using the images in the second row of Figure 6.6. These images have the shadows on different zones around the border of the human figure. The reconstruction in Figure 6.8 shows that the human shape is well computed and furthermore, that the flat region around this shape has been well preserved.

7. Conclusions. This study utilized numerical schemes commonly employed in the Shape from Shading literature also for the PS_2 problem under the perspective projection assumption. We proved the uniqueness of the solution in the class of Lipschitz continuous surfaces given Dirichlet boundary conditions. We then extended the two numerical methods of Mecca and Falcone [22], the upwind finite difference scheme and the semi-Lagrangian scheme, for the solution of the 2-image *perspective* Photometric Stereo problem. As the method of Mecca and Falcone [22, 20] can also reconstruct the albedo in a manner similar to the suggested perspective one, the inaccurate orthographic reconstruction is not due to the non-constant albedo, but rather a result of the unrealistic set of assumptions of a perspective projection in the proposed algorithms.



(original) (up-wind) (semi-Lagrangian)

Figure 6.5. *All the surfaces shown here are related to the 2000×2000 pixels images with* 5% *of Gaussian noise.*



I_1 I_2 I_3

Figure 6.6. *Images of the Beethoven bust and coin with occlusions.*

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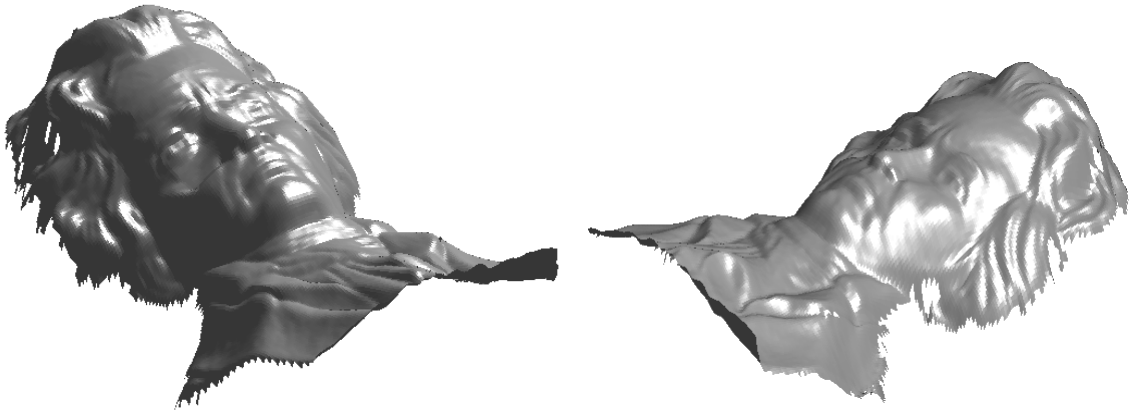


Figure 6.7. *Novel views of the reconstruction of Beethoven using W-PS_3.*

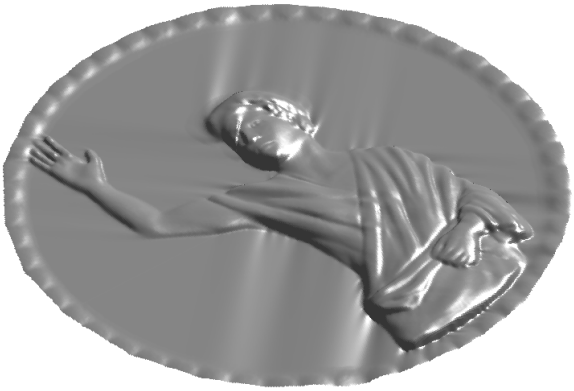


Figure 6.8. *A novel view of the reconstruction of a coin model using W-PS_3.*

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